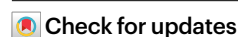


# A wireless modular platform for neuro-behavioral recording and closed-loop manipulation in small animals

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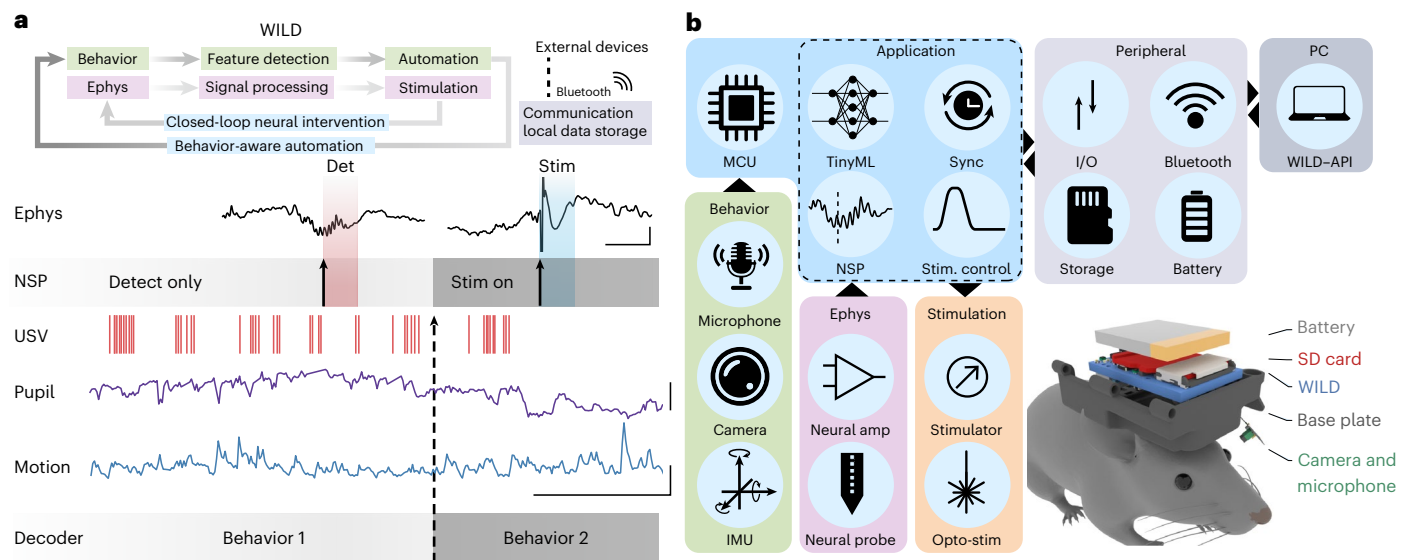
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Investigating the neural mechanisms of social group interactions and other naturalistic behaviors in small animals remains limited by current technology. Tethered neural recording systems are incompatible with many of these behaviors, while existing wireless devices for small animals are constrained by weight, bandwidth, recording duration and the lack of closed-loop modulation capabilities. Here we overcome these limitations by developing a Wireless, Interactive, Lightweight Datalogger (WILD) with integrated flexible neural probes, optogenetics, an inertial measurement unit, an ultrasonic microphone and a head-mounted camera. This platform enables simultaneous, long-term recording of neural activity, locomotor variables, vocalizations and eye movements from groups of freely moving mice in both laboratory and outdoor settings. Model-based real-time signal processing detects specific neural events and behavioral motifs to trigger closed-loop neural interventions. By combining multimodal recordings with advanced onboard signal-processing capabilities in a compact device, WILD enables the investigation of neural mechanisms underlying a broad range of natural behaviors in small animals.

A fundamental goal in neuroscience is to uncover the neural mechanisms of behavior. To achieve this goal, it is necessary to both record and precisely manipulate the activity of populations of neurons and relate it to quantitative descriptions of behavioral dynamics. Remarkable progress has been made in recent years in the development of tools for large-scale neural recordings<sup>1,2</sup>, manipulating the activity of specific cell types with high spatiotemporal resolution<sup>3,4</sup> and accurately tracking behavioral and physiological variables<sup>5–7</sup>, particularly in laboratory rodents. However, owing to technical limitations, these methods have been mostly deployed in reduced and simplified behavioral paradigms that fail to fully capture the richness and complexity of natural behaviors. As a result, there is an increasing need to expand the behavioral repertoire of mechanistic neuroscience studies to include more naturalistic and ecologically relevant behaviors<sup>8,9</sup> such as social group dynamics, navigation and foraging in large and complex environments, and even going beyond the laboratory to outdoor settings<sup>10–12</sup>.

This need has led to a growing demand for approaches that enable mechanistic studies of neural activity in the context of unrestricted, naturalistic behaviors.

The most common methods for recording neural activity in freely behaving animals involve the use of tethered electrophysiological or imaging devices. These approaches enable the simultaneous recording of large numbers of neurons, but at the cost of restricting animals' natural movements, the size and complexity of the behavioral apparatuses and the range of behaviors (for example, group interactions). Wireless recording methods enable neural recordings during unrestrained behavior; however, they currently have substantial limitations. Battery weight and energy efficiency limit recording duration and electrode numbers, especially in small animals such as mice. Existing wireless neural recording interfaces are designed without considering the need for broader functionality, such as manipulating neural activity or incorporating other types of sensors to record behavioral and physiological



**Fig. 1 | A wireless platform for recording and control of neural and behavioral activity in freely behaving small animals.** **a**, Top: a schematic of WILD operations. Behavioral and electrophysiological signals are processed onboard to support feature detection, behavior-aware automation and closed-loop neural intervention. Data are stored locally, while wireless communication enables synchronization and monitoring through a remote device. Bottom: WILD enables wireless recording of neural electrical activity (electrophysiology (Ephys); scale bar, 0.5 mV, 50 ms) alongside behavioral variables, including USVs, pupil dynamics and locomotor parameters (scale bars, pupil diameter (Pupil) 0.2 a.u.; locomotion speed (Motion) 2 cm s<sup>-1</sup>, 5 s), in freely behaving mice.

**b**, A system diagram of WILD. Behavioral and electrophysiological signals are fed into TinyML and NSP modules and stored in a microSD card. Real-time detection (Det) of neural events and behavioral motifs is performed on board and can be used to control optogenetic manipulations in a closed-loop manner (Stim). Communication with the host computer is achieved via Bluetooth, and seamless synchronization with external devices is provided, all controlled by WILD-API. The inset illustrates a mouse implanted with a WILD device, electrodes, battery and protective 3D casing. Amp, amplifier; ephys, electrophysiological recording; opto-stim, optogenetic stimulation.

variables<sup>13–16</sup>. Another key limitation of current wireless neural systems is their inability to perform complex onboard signal processing, which limits their application in studies requiring closed-loop neural intervention. Recent advances in machine learning (ML), particularly in developing lightweight model frameworks on embedded processors, offer a pathway toward overcoming this limitation<sup>17</sup>. Embedding efficient ML algorithms and artificial neural networks (ANNs) could, in principle, endow wireless neural interfaces with advanced capabilities for real-time detection and manipulation of neural or behavioral events, but this application is yet to be developed.

To enable mechanistic studies of fully unrestricted natural behaviors in small animals, we developed the Wireless, Interactive, Lightweight Datalogger (WILD), an integrated platform for wireless recording of neural activity and multiple behavioral and physiological variables (Fig. 1a). Embedded real-time signal processing and efficient ANN architectures enable wireless closed-loop optogenetic manipulations based on real-time detection of neural and behavioral events (Fig. 1a). We demonstrate the capabilities of this system in sets of experiments not feasible with currently existing approaches, including recordings of neural activity and behavioral variables (such as vocalizations, pupil dynamics and locomotion) during social group behavior of mice, and neural and behaviorally triggered closed-loop manipulations in mice navigating complex three-dimensional (3D) environments. To facilitate the adoption of this system across different laboratories, all hardware and software elements of this platform are open-source and fully documented.

## Results

### A light-weight, multimodal, neural and behavioral datalogger

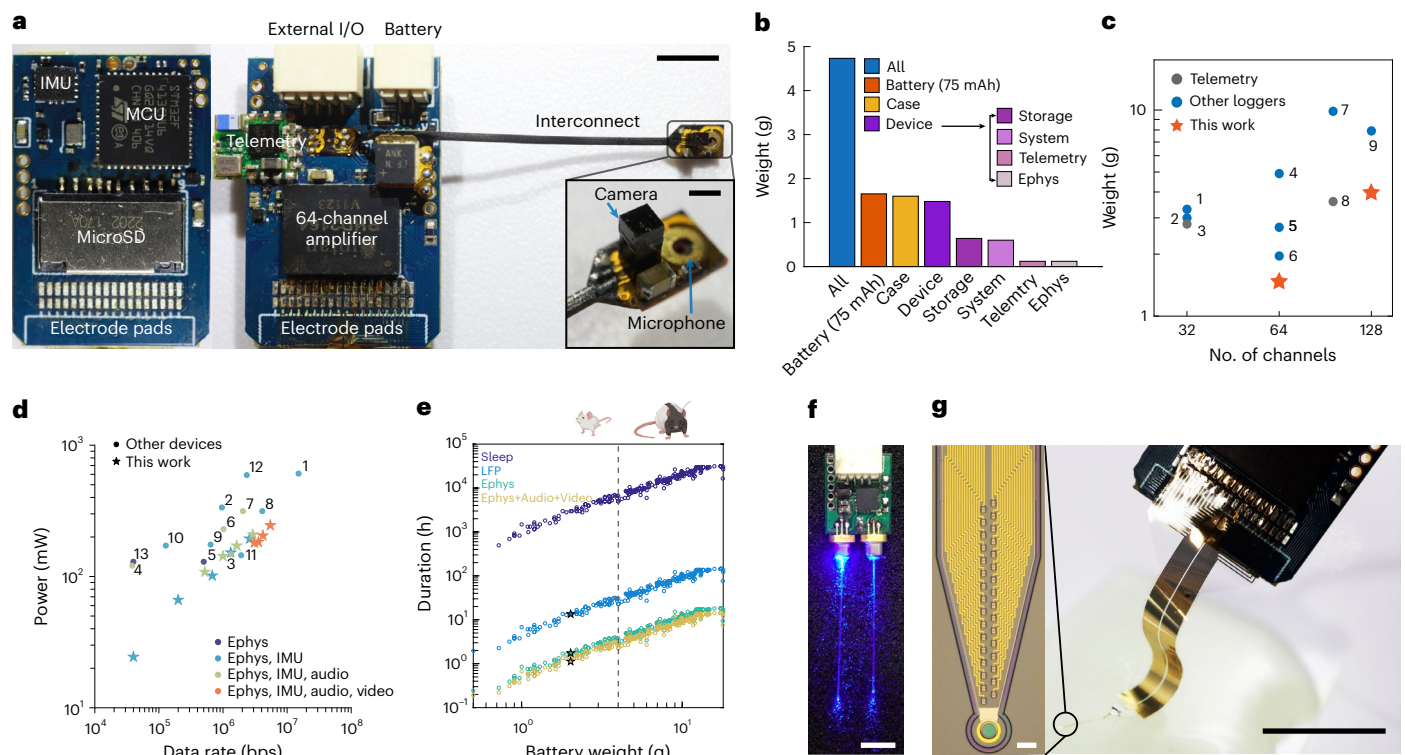
The core element of our platform is a light-weight multimodal datalogger (Fig. 1b and Extended Data Fig. 1a,b). While wireless neural recording systems have been previously developed<sup>18,19</sup>, our device overcomes existing barriers that have limited the broader adoption of

these technologies and enables experimental paradigms that integrate multimodal sensing and onboard ML within an open-source framework.

Traditional wireless neural interfaces have usually been limited to the recording of neural activity, lacking other important functions, such as multimodal sensing or on-demand manipulation. Our platform's basic module features a 64-channel neural amplifier, a microSD card for data storage and a Bluetooth module for wireless data communication up to 70 m (Fig. 2a and Extended Data Fig. 1c). In addition, it includes additional sensors for monitoring behavioral variables. A nine-axis inertial movement unit (IMU) enables high-resolution motion tracking, a 320 × 320 pixel micro-camera captures pupil dynamics or animal point of view and an ultrasonic microphone records animal vocalizations (Fig. 2a). All these different data streams are simultaneously recorded and seamlessly synchronized together with external devices such as video cameras, through our open-source WILD application programming interface (WILD-API), including an open-source graphical user interface and command-line interface (Extended Data Fig. 1d,e). We also provided documents for basic device operation command through Bluetooth low-energy traffic (Supplementary Table 2).

Including these expanded functionalities, the total weight of the device is below 1.5 g, in a 23.3 mm × 15.7 mm circuit board (Fig. 2b). Compared to existing wireless neural recording and telemetry systems (that lack these advanced sensing and ML capabilities), WILD achieved the lowest weight (Fig. 2c), facilitating its use in freely moving small animals such as mice or birds.

Another key element of WILD is a compact high-performance microcontroller unit (MCU; Extended Data Fig. 1f). We developed an efficient signal acquisition pipeline that handles all data internally without the need for external memory or data processing modules, reducing device footprint and power consumption. As a result, WILD can record for longer periods of time than previous wireless recording systems (normalizing by channel count and battery capacity) (Fig. 2d,e



**Fig. 2 | Device characterization.** **a**, A picture of the WILD basic module (scale bar, 5 mm). The inset shows a zoomed-in picture of a multimedia capture module with embedded micro-camera and microphone (scale bar, 1 mm). **b**, A weight breakdown of the different components that integrate the device, including the battery. The device itself weights <1.5 g, reaching ~4.5 g with the case and battery. **c**, A weight versus channel count comparison of WILD (stars) with existing wireless neural interfaces<sup>13,19,48–50</sup>. See Supplementary Table 1 for detailed comparisons. **d**, Power efficiency (consumption versus data recording rate) comparison of different versions of the recording modes of WILD (stars) with existing wireless neural interfaces (dots; labels correspond to the reference number)<sup>13,19,48–50</sup>. **e**, Estimated operating times under power-saving idle mode

for sleep, LFP (1,250 Hz), only Ephys (20 kHz) and Ephys (20 kHz) with audio and video recording, using various lithium-ion batteries. The vertical dashed line indicates the approximate weight limit for freely moving mice (~5 g, including the battery). Note that the system provides ~20 h of recording at 1,250 Hz or ~4 h at 20 kHz. The stars denote values measured with a 90-mAh battery. **f**, A picture of the stimulation module with mounted laser diodes and optic fibers (scale bar, 5 mm). The weight of this module is 0.6 g. **g**, The implant-ready device with a mounted flexible neural probe. Left: a microscopic image of high-density paralyne-C flexible neural probe (scale bar, 20  $\mu$ m). Right: a photo showing the mounted probe flattened on a phantom brain (scale bar, 10 mm). Ephys, electrophysiology.

and Extended Data Fig. 1g–i). For example, a ~2.5-g lithium-polymer battery (130 mAh), that can be comfortably carried by a mouse, enables up to ~3 h of 64-channel recordings at 20 kHz (typical sampling rate for recording extracellular neuronal spikes) (Fig. 2e). For larger animals, such as rats, a battery of ~5 g would enable ~9 h of recording (64 channels at 20 kHz; Fig. 2e). In addition, the MCU incorporates embedded signal processing and ML algorithms for real-time neural and behavioral event detection and generates trigger signals to control closed-loop optogenetics or electrical stimulation.

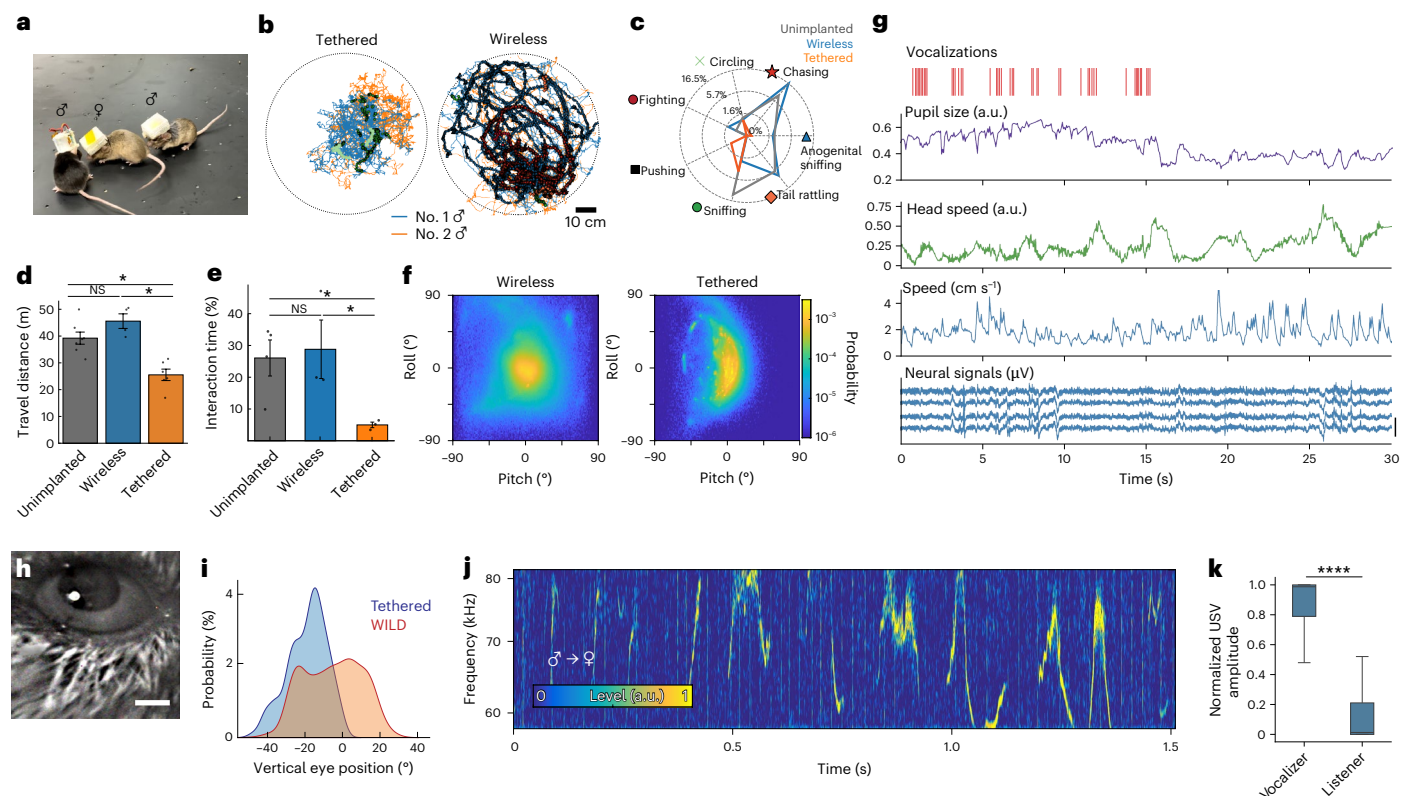
In addition to its embedded sensors, WILD is a modular, customizable platform that supports the expansion of channel counts (Extended Data Fig. 2a) or integration of additional modules through analog and digital interfaces (Extended Data Fig. 2b). An example of this is a miniaturized current stimulation module with integrated laser diodes and optic fibers for onboard optogenetic manipulations (Fig. 2f and Extended Data Fig. 2c). The combination of this stimulation module with WILD's onboard real-time signal processing capabilities enables optogenetic stimulation triggered upon the detection of neural or behavioral patterns with sub-millisecond latency (Extended Data Fig. 2d,e), allowing precise real-time manipulations. All these functions can be flexibly configured through the WILD-API, enabling adaptable system operation for different experimental requirements (for example, arbitrarily defined stimulation protocols or triggering with different neural or behavioral events).

Finally, WILD devices implanted in multiple animals can be synchronized through Bluetooth communications. Millisecond-precision

synchronization was enabled by an onboard precision crystal oscillator and a two-way time-calibration process to ~1-ms precision (Extended Data Fig. 2f,g). External systems could be synchronized through an expansion inputs/outputs (I/O) interface board (Extended Data Fig. 2h). This setup enables recording from multiple animals simultaneously, with all WILD devices and external input signals synchronized together.

WILD can interface with different common types of commercially available or custom electrodes through standard connectors (Extended Data Fig. 3a). To achieve maximal compactness and minimal weight with high-channel count electrodes, we microfabricated flexible neural probes on the basis of a flexible paralyne-C substrate and directly bonded them to the WILD board (Fig. 2f and Extended Data Fig. 3b). Each probe can be implanted independently, allowing the targeting of multiple brain regions simultaneously (Extended Data Fig. 3c). Moreover, these flexible probes eliminate mechanical mismatches between probe and tissue, enabling stable long-term single-unit recordings. Directly bonding them to the pads in the WILD circuit board eliminates the need for additional connectors, enabling the probes, datalogger and battery to fit inside a small 3D-printed casing (Extended Data Fig. 3d–f).

In summary, we have developed a fully integrated wireless, modular platform for energy-efficient monitoring of neural activity and behavioral variables, enabling real-time signal processing and closed-loop optogenetic manipulations in freely moving small animals such as mice.



**Fig. 3 | Multimodal wireless recordings with WILD during mice's unrestricted group social behavior.** **a**, A picture of socializing mice implanted with WILD. The circle with a cross below it represents female mice; the circle with an arrow pointing up and to the right represents male mice. **b**, Trajectories of two mice during free social interaction on an open maze during both tethered (left, using OpenEphys system) and wireless (right, using WILD) recording. Orange and blue traces represent the movements of two individual males. Different symbols mark the occurrence of different behaviors corresponding to **c**. **c**, A diagram showing fraction of time of male mice with different recording conditions engaged in different social behaviors. **d**, The average travel distance of unimplanted (gray,  $n = 7$  sessions from two mice), wireless (blue,  $n = 4$  sessions from two mice) and tethered recorded mice (orange,  $n = 6$  sessions from two mice). Unimplanted versus wireless:  $P = 0.109$ , Mann–Whitney  $U$  test,  $P = 0.109$ , Neniadini–Hochberg FDR; unimplanted versus tethered:  $P = 0.0023$ , Mann–Whitney  $U$  test,  $P = 0.007$ , FDR; wireless versus tethered:  $P = 0.0095$ , Mann–Whitney  $U$  test,  $P = 0.0143$ , FDR. Data are presented as mean  $\pm$  s.e.m. **e**, Male–male interaction time as a percentage of total session duration across different recording

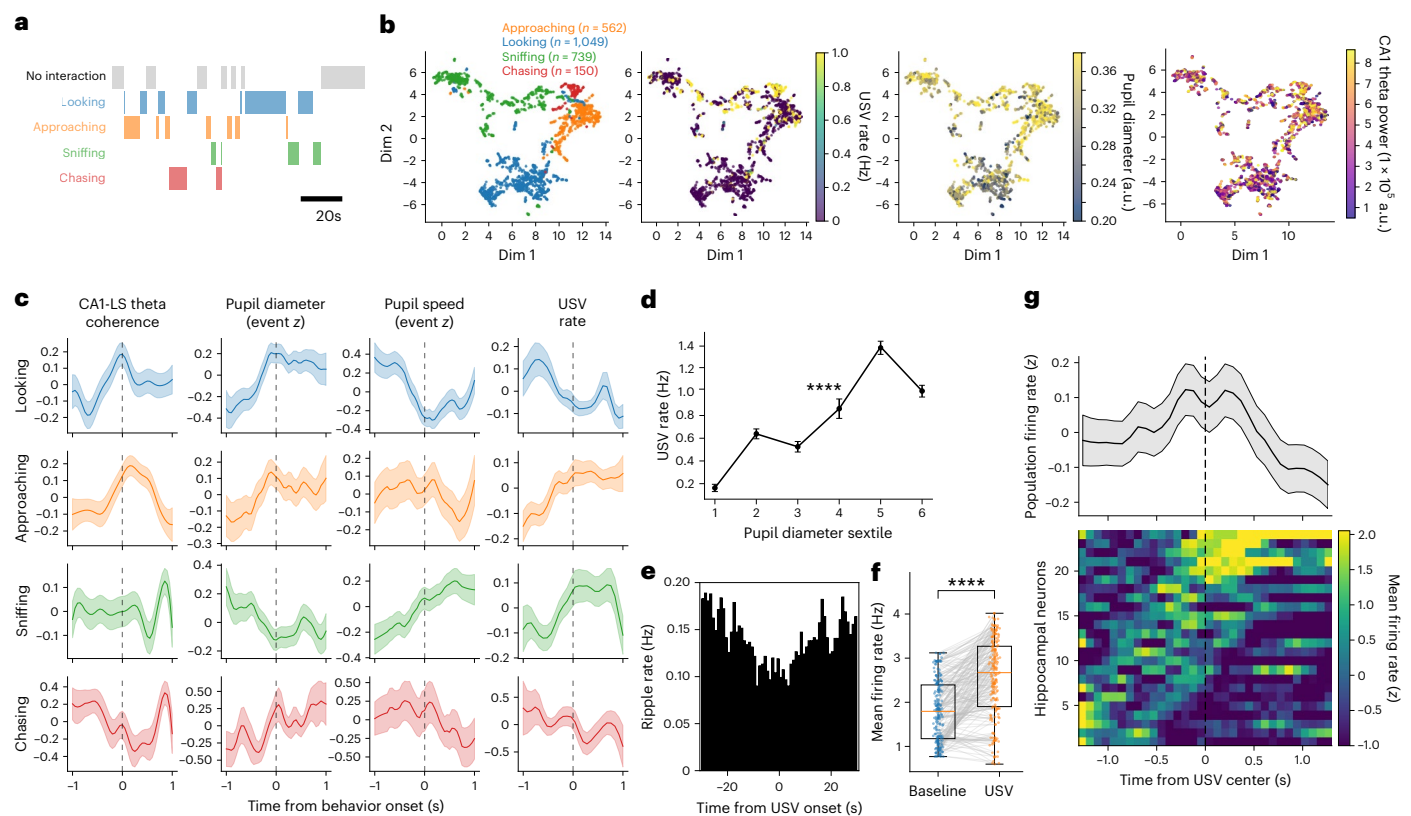
conditions. Unimplanted versus wireless:  $P = 0.62$ , Mann–Whitney  $U$  test,  $P = 0.62$ , Neniadini–Hochberg FDR; unimplanted versus tethered:  $P = 0.0114$ , Mann–Whitney  $U$  test,  $P = 0.0342$ , FDR; wireless versus tethered:  $P = 0.0249$ , Mann–Whitney  $U$  test,  $P = 0.0374$ , FDR,  $n = 10$  sessions). Data are presented as mean  $\pm$  s.e.m. **f**, Comparison of head pitch and roll between tethered and wireless mice (roll/pitch: wireless versus tethered  $P = 2.5527 \times 10^{-134}$ ,  $P < 10^{-100}$ , Mann–Whitney  $U$  test,  $n = 10$  sessions). **g**, An example of multimodal signals recorded from socially interacting mice. Scale bar, 1,000  $\mu$ V. **h**, A mouse pupil captured by integrated micro-camera (scale bar, 50 pixels). **i**, The distribution of vertical eye position during behavior (Kolmogorov–Smirnov test:  $D = 0.43$ ,  $P = 1.54 \times 10^{-82}$ ; Levene's test for variance:  $F = 198.4$ ,  $P = 5.97 \times 10^{-43}$ ,  $n = 10$ ). **j**, An example spectra of male mouse USVs recorded with onboard microphone during social interaction with a female mouse. **k**, The relative amplitude for the same call as recorded by the microphone on the vocalizer and listener devices ( $n = 969$  USVs,  $P < 5.769 \times 10^{-180}$ , Wilcoxon test). The box plots show median and 25th and 75th percentiles, with whiskers as the minimum and maximum values. In **d**, **e** and **k**,  $*P < 0.05$ ,  $****P < 0.0001$ .

### WILD enables multimodal recordings during mouse social group behaviors

Mice are social animals that, in their natural habitats, live in groups with complex social dynamics<sup>20,21</sup>. However, the neural mechanisms of social behaviors in groups of freely interacting mice have been barely studied owing to the limitations of current technologies. Tethered recordings severely restrict behaviors such as courtship or aggression and cannot be applied to multiple freely interacting animals owing to cable entanglement (Supplementary Video 1). Moreover, previous wireless neural interfaces have been limited in capturing behavioral signals together with high-bandwidth neural data, and have not been able to deliver responsive neural stimulation. To showcase how WILD enables experiments not feasible with previous approaches, we recorded groups of two to four male and female mice freely interacting in a large arena. Each mouse was implanted with a WILD module interfacing with a flexible neural probe and connected with a lithium-polymer battery (Fig. 3a).

Compared to tethered conditions, mice implanted with WILD were more active, covered a larger area of the maze, and showed a

larger spectrum of social behaviors, including sniffing, circling, chasing and fighting (Fig. 3b–e and Supplementary Video 1). The percentage of time spent performing different social behaviors was similar between WILD-implanted and unimplanted mice, whereas tethered mice spent most of their time pushing away conspecifics (Fig. 3c,e). WILD-implanted mice traveled longer distances and displayed a larger variety of social behaviors and prolonged interaction times compared to tethered mice, similar to unimplanted animals (Fig. 3b–e and Extended Data Fig. 4a–d). The onboard nine-axis IMU enabled 3D reconstruction of head movements for video-free behavior tracking (Extended Data Fig. 4e–g), which is particularly relevant during behaviors such as conspecific sniffing and fighting. WILD-implanted mice displayed a wider range of head pitch and roll than tethered mice, while mostly maintaining a central position not disputed by tether drag (Fig. 3f, Extended Data Fig. 4h). In addition, WILD-implanted mice showed significantly higher angular head mobility than tethered mice, suggesting WILD poses less head movement constraints ( $P = 0.032$ , two-sided Kolmogorov–Smirnov test; Extended Data Fig. 4i).



**Fig. 4 | WILD reveals physiological correlates of different social behaviors.**

**a**, An ethogram showing distinct types of interactions during an example male-male interaction session. **b**, Left: two-dimensional UMAP of body pose and IMU features in a representative male during male-female interactions. The color code represents manually annotated behaviors: approaching (orange,  $n = 562$ ), looking (blue,  $n = 1,049$ ), sniffing (green,  $n = 739$ ) and chasing (red,  $n = 150$ ). Right: same embedding colored by USV rate, pupil diameter and hippocampal CA1 theta power. **c**, Peri-event modulation of neural and behavioral variables (columns) triggered by different social behaviors (rows) in males. Traces show the mean  $\pm$  s.e.m. from  $-1$  to  $1$  s around behavior onset ( $0$  s, dashed line) for looking ( $n = 70$ ), approaching ( $n = 88$ ), sniffing ( $n = 108$ ) and chasing ( $n = 14$ )

events ( $n = 2$  mice). **d**, Mean  $\pm$  s.e.m. USV rate (Hz) is shown for each sextile of pupil diameter ( $r = 0.131$ ,  $P = 1.47 \times 10^{-40}$ ,  $n = 4,949$  USVs from six sessions and three mice). **e**, A peri-event histogram of hippocampal ripple rate aligned to USV onset (stats and  $n$ ). **f**, Average CA1 firing rate during USVs, restricted to low-speed noninteraction periods ( $< 4 \text{ cm s}^{-1}$ ), and matched sample periods of non-USV (baseline) times ( $P = 3.73 \times 10^{-19}$ , Wilcoxon signed-rank test,  $n = 201$  pairs,  $n = 24$  neurons). The box plots show median and 25th and 75th percentiles, with whiskers as the minimum and maximum values. **g**, Hippocampal CA1 neuron firing response around USVs ( $n = 201$  events). Top: the mean  $\pm$  s.e.m. population firing rate. Bottom: a raster plot of individual neurons. In **d** and **f**, \*\*\*\*  $P < 0.0001$ . Dim, dimension.

A key feature of WILD is that, in addition to neural recordings, it enables simultaneous acquisition of multiple behavioral variables (Fig. 3g). For example, mice emit different types of ultrasound vocalizations (USVs) during social interactions<sup>22,23</sup>. Pupil dynamics reveal changes in attention, arousal and brain state<sup>24</sup>, but they are difficult to monitor in the context of social interactions. WILD onboard microphone and camera allowed us to record USVs and pupil dynamics with high resolution during unrestricted social group behaviors (Fig. 3g). Our miniaturized head-mounted camera did not interfere with mice's vision or eye movements, as indicated by their wider range of eye movements compared to recordings performed with previously developed, tethered, head-mounted cameras<sup>24-26</sup> (Fig. 3h,i and Extended Data Fig. 5a-e). WILD ultrasound microphone allowed for the detection of USVs during male-female interactions (Fig. 3j). Head-mounted microphones allowed us to differentiate between self-generated and conspecific USVs during unrestricted social interactions, something not feasible with overhead external microphones (Fig. 3 and Extended Data Fig. 5f-k). To verify this, we took advantage of the fact that during male-female interactions, almost all USVs are produced by the male<sup>27</sup>. We found that most USVs showed high amplitude in the vocalizer (male) head-mounted microphone channel and near-zero amplitude in the listener (female) microphone channel (Fig. 3k). These simultaneous multimodal recordings could thus enable dissection of the behavioral and neural basis of group social behaviors in mice.

To showcase the types of physiological insights that can be uncovered with this approach, we implanted mice with WILD devices and simultaneous flexible probes in the hippocampus and lateral septum (LS), important regions for social behaviors<sup>28</sup>. During unrestricted male-female interaction sessions in an arena, mice engaged in a range of different social behaviors (Fig. 4a). To visualize the heterogeneity of male behavior, we extracted features of animal pose (derived from video) and WILD IMU, embedded them in a low-dimensional representation and color-coded each interaction according to manually verified labels (Fig. 4b, left). Projecting USV rate, pupil diameter and hippocampal CA1 theta-band power ( $5-10 \text{ Hz}$ ) onto the same uniform manifold approximation and projection (UMAP) embedding revealed that these multimodal physiological and neural variables were differentially associated with distinct social behaviors (Fig. 4b). To explore this relationship in a more quantitative manner, we analyzed hippocampal CA1-LS oscillatory theta coherence, pupil diameter and speed, and USV rate around the onset of different behaviors (Fig. 4c and Extended Data Fig. 6). We found that 'looking' behavior was associated with increased CA1-LS theta coherence, larger pupil diameter and reduced pupil speed, suggesting relatively stable gazing, as well as a lower rate of USVs (Fig. 4c). During 'approaching' behavior, CA1-LS theta coherence increased, along with pupil diameter and USV rate. However, during 'sniffing', pupil diameter decreased whereas pupil speed increased accompanied by a higher USV rate. Finally, during

'chasing', we found a reduced CA1-LS theta coherence despite increased pupil diameter (Fig. 4c).

Overall, USV rate was strongly correlated with pupil size (Fig. 4d), suggesting that vocalizations are generated during transient states of increased arousal. We observed that most USVs coincided with periods of immobility or low-speed locomotion (Extended Data Fig. 6). During these periods, hippocampal activity is dominated by sharp-wave ripple (SWR) oscillations that strongly entrain neuronal activity. Surprisingly, USVs and SWRs were anticorrelated (Fig. 4e and Extended Data Fig. 6). Instead, hippocampal activity was selectively enhanced during USV events: CA1 neurons showed higher firing rates during USVs than during matched non-USV immobility periods (Fig. 4f). Furthermore, a subset of hippocampal neurons (~40%) were strongly recruited during USV generation (Fig. 4g). Together, these results suggest that USV generation is associated with a distinct hippocampal activity state during immobility, distinct from SWRs, that selectively engages a subset of CA1 neurons.

### Neural and behavior recordings of groups of mice living in outdoor enclosures with WILD

Rodents in their natural habitats navigate and forage for food over long distances in complex environments. However, the neural mechanisms underlying these behaviors are typically studied in reduced laboratory mazes whose dimensions and complexity are limited by the use of tethers. To demonstrate the capabilities of the WILD system for the study of natural behaviors, we recorded rats and mice in very large outdoor enclosures (Fig. 5 and Extended Data Fig. 7). During fabrication, the WILD board was coated by a thin film of parylene-C, protecting it from humidity and mechanical damage, enabling animals to carry the device for multiple days while living outdoors, without compromising its function (Fig. 5a–e).

In one of these experiments, we implanted nine male C57 mice with silicon probes in the dorsal hippocampus and WILD devices. The mice lived in an outdoor enclosure (38 m × 15 m) for 15 days (Fig. 5a,b). Food was placed in nine specific resource zones in the environment (wooden 'shelters'; Fig. 5a). To investigate how mice explore a novel natural environment and characterize their spatial, social and circadian behaviors and their neural correlates, we tracked their behavior and recorded neural activity using WILD. Animal behavior was continuously recorded during this time using multiple cameras around the enclosure as well as an ultrawide band (UWB) position tracking module attached to the main WILD board (Fig. 5c). Neural recording quality and noise levels were not different from those obtained in the laboratory with standard tethered recordings (Fig. 5d and Extended Data Fig. 8), and remained stable while mice were living outdoors (Fig. 5e).

We first characterized mice movement patterns. There was a strong circadian modulation with mice becoming increasingly active shortly before sunset and remaining highly active until sunrise.

(Fig. 5f,g). On average, mice traveled  $6.20 \pm 0.71 \text{ m h}^{-1}$  during the day and  $9.56 \pm 0.9 \text{ m h}^{-1}$  during the night (Fig. 5g). Conversely, mice spent a large fraction of the day sleeping, with long bouts of non-rapid eye movement (NREM) sleep (Fig. 5h,i). During NREM sleep, hippocampal activity is dominated by SWRs, synchronous network oscillations that support memory formation<sup>29–31</sup>. SWR rate displayed a surprising circadian modulation. It was relatively low during the day, despite the higher proportion of NREM sleep, and increased significantly during the night ( $P = 0.0066$ , Wilcoxon rank-sum test; Fig. 5j,k). This observation suggests an enhanced memory consolidation during brief periods of sleep during nighttime, perhaps due to an increased amount of new information gathered during those periods.

During navigation, hippocampal 'place cells' fire at specific locations, encoding as a population an internal 'map' of the environment<sup>32</sup>. We were able to isolate and classify a large number of single neurons from these recordings (Fig. 5e and Extended Data Fig. 7) and identify multiple place cells. Place cells in the outdoor enclosure exhibit similar characteristics to those observed in much smaller laboratory mazes (Fig. 5l–o). Place field properties were heterogeneous across the population, with some cells exhibiting multiple fields and others single ones (Fig. 5l), while the distribution of field size, spatial information and selectivity displayed a heavy-tailed distribution across the population (Fig. 5m–o).

Overall, this data demonstrate the utility of WILD to study the neural mechanisms of natural behaviors in rodents and other small animals (Fig. 5).

Despite its energy-efficient design, WILD can only record at high bandwidth (64 channels, 20 kHz) continuously for ~3–9 h (depending on battery capacity; Fig. 2e), although recordings can be considerably longer at lower sampling rates (Extended Data Figs. 1 and 7). In some cases, it may be required to investigate behaviors that unfold over longer periods of time, during which retrieving animals to change batteries is not desirable. To support long-term, on-demand recording with minimal disturbance to the animal, we have designed the system with highly scalable power modes. When WILD is in idle mode, it is powered with a minimal power state (~1 mW), which could last for weeks with a 1.5-g battery (80 mAh, ~300 h). Switching between high-power (for 20 kHz, 64-channel recording) and low-power modes can be done by sending a remote command to the WILD module via Bluetooth or it can be triggered by specific device events. This enables restricting recordings to specific periods of active behavior, while saving battery life during prolonged periods of immobility and sleep.

### Onboard real-time signal processing and TinyML enable neural and behavioral closed-loop manipulations

A powerful approach to investigate the neural mechanisms of behavior is to test the functional role of specific neural patterns or behavioral

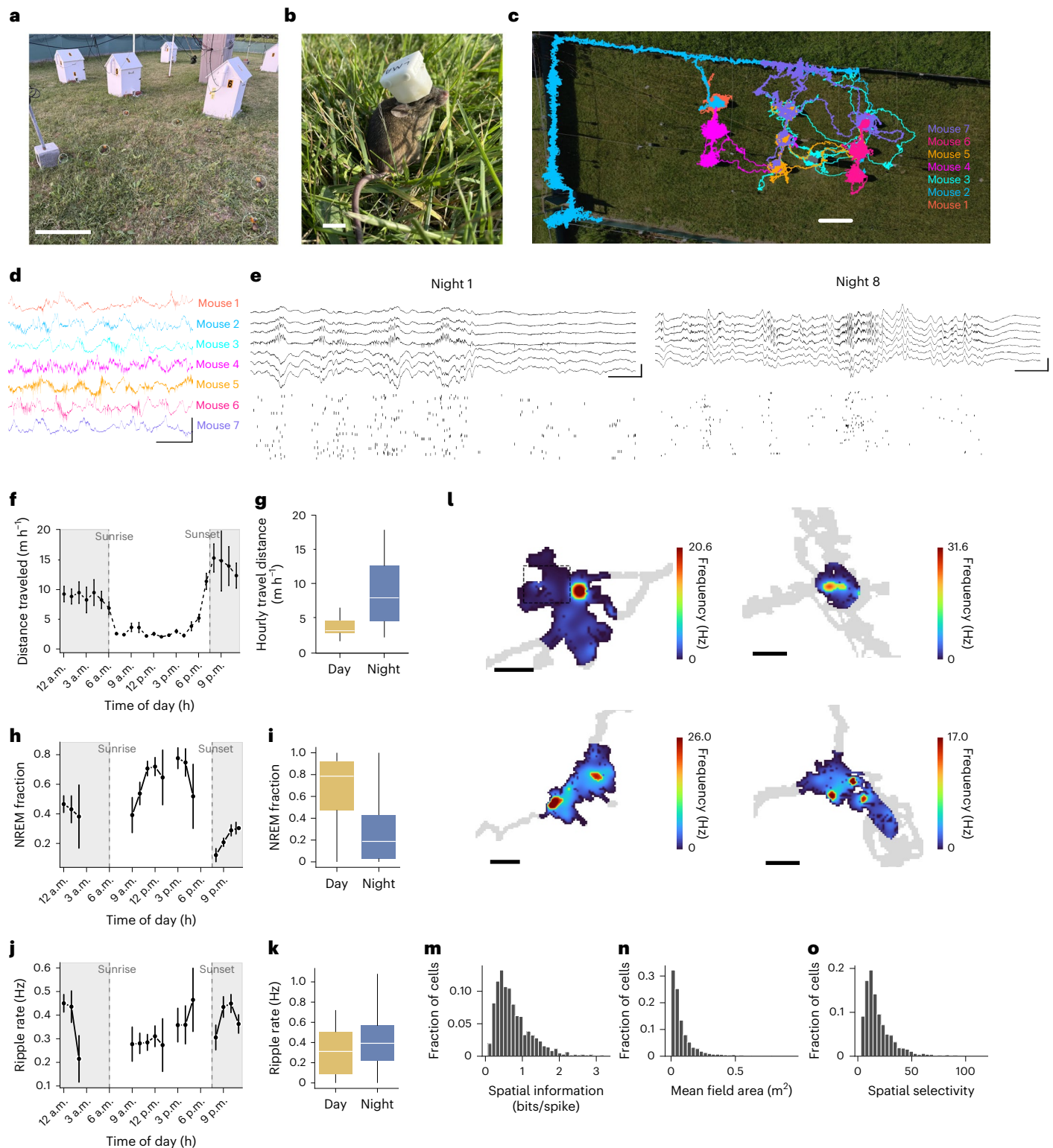
**Fig. 5 | Wireless electrophysiology in freely moving mice in naturalistic outdoor settings.** **a**, A view of the semi-natural field enclosure with resource zones (white boxes). Six mice implanted with WILD devices are visible in the grass (white circles). Scale bar, 1 m. **b**, A close-up of a mouse implanted with a silicon probe and WILD. Scale bar, 1 cm. **c**, The trajectories of seven simultaneously recorded mice on night 6 from 7 p.m. to 7 a.m. the next day. Each color represents a different animal. Movements were recorded using WILD's UWB module. **d**, Simultaneous hippocampal LFP traces from the same seven mice. Scale bar, 1 mV, 200 ms. **e**, Representative hippocampal LFP traces (top) and single-neuron rasters (bottom) from the same animal on night 1 (left) and night 8 (right), demonstrating recording stability of WILD. Scale bar, 1 mV, 100 ms. **f**, Mean distance travel per hour across a 24-h cycle ( $n = 9$  mice, nights 1–15). The gray shading indicates nighttime. Data are presented as mean  $\pm$  s.e.m. **g**, Distance traveled per hour was significantly greater during the night than during the day (day:  $6.20 \pm 0.71 \text{ m h}^{-1}$ ; night:  $9.56 \pm 0.91 \text{ m h}^{-1}$ ;  $n = 9$  animals;  $P = 1.1 \times 10^{-6}$ , Wilcoxon rank-sum test). **h**, Fraction of time spent in NREM sleep across the

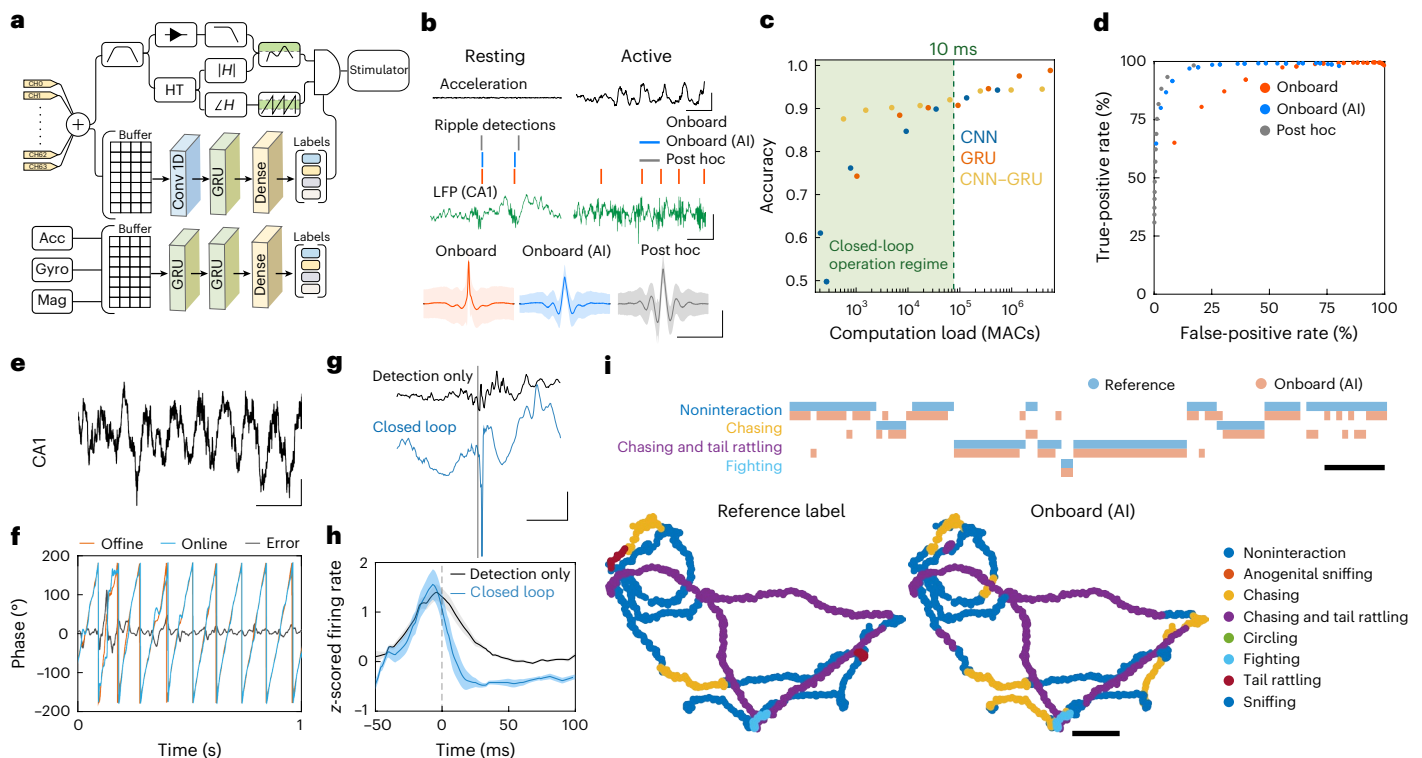
day ( $n = 9$  animals, 15 days). Data are presented as mean  $\pm$  s.e.m. **i**, Day versus night NREM fraction. ( $n = 9$  animals,  $P = 1.48 \times 10^{-17}$ , Wilcoxon rank-sum test). **j**, Mean SWR rate during NREM sleep (Hz) across the day ( $n = 9$  animals). **k**, Day versus night NREM SWR rate ( $n = 9$  animals,  $P = 0.0066$ , Wilcoxon rank-sum test). **l**, Example rate maps from four representative hippocampal place cells recorded in different animals in the enclosure. Examples were chosen to illustrate the heterogeneity of hippocampal place cells in the enclosures (for example, with single or multiple place fields). The dashed boxes denote resource zones. Scale bar, 0.5 m. **m**, The distribution of spatial information (bits/spike) for all identified place cells ( $n = 1,636$  cells from nine animals). **n**, The distribution of mean place field area ( $\text{m}^2$ ) for all identified place cells ( $n = 1,636$  cells from nine animals). **o**, The distribution of spatial selectivity (peak in-field/mean firing rate) for all identified place cells ( $n = 1,636$  cells from nine animals). The box plots show median and 25th and 75th percentiles, with whiskers as the minimum and maximum values.

motifs by detecting them in real time and delivering rapid, on-demand manipulations, such as optogenetic stimulation. This ‘closed-loop’ approach offers more precise control of neural circuit operations and enables linking specific neural patterns to behavioral outcomes<sup>33,34</sup>. Despite its advantages, its integration with wireless recording methods has been prevented by the challenge of real-time signal processing and stimulation delivery in embedded systems. Moreover, recent breakthroughs in the study of animal behavior have been catalyzed by the application of ML algorithms for the unsupervised classification of behavioral motifs, using either video<sup>5,6</sup> or IMU<sup>33,35</sup> data. However, these

methods have not yet been deployed in embedded systems, preventing their integration with wireless platforms.

We tackled these challenges by developing a suite of real-time signal processing algorithms and efficient, light-weight ML models that can be executed directly by WILD’s onboard processor. Both neural and behavioral signals (IMU, sound and video) can be taken as inputs, enabling the rapid detection of transient neural activity patterns, specific behavioral motifs or a combination of both. We have implemented three parallel onboard signal processing pipelines for detecting neural and behavioral events in real time. The first pipeline





**Fig. 6 | Neural and behavioral closed-loop manipulations with WILD in behaving mice.** **a**, Schematics of embedded real-time signal processing. Neural signals are processed by a low-latency digital signal processing pipeline, including a Hilbert-transform (HT)-based envelope and phase detector derived from the analytic signal ( $H$ ) (top). Neural signals are also cached and analyzed by an embedded TinyML model (bottom). Outputs from these pipelines are used to control stimulation. **b**, An example of SWR detection in a mouse during sleep (left) and behavior (right). The vertical lines indicate real-time SWR detection with only onboard signal processing (orange), with added TinyML denoising (blue), and post hoc (gray). The green trace is hippocampal LFP. Note the spurious SWR detection during behavior due to EMG noise (right). The graphs at the bottom show average traces of detected events. Scale bars, 100 ms, 1 g (top); 100 ms, 0.5 mV (middle); 50 ms, 100  $\mu$ V (bottom). **c**, The estimated accuracy of TinyML noise identification model compared to model size and structures. Target inference times are indicated by a dashed line. **d**, Receiver operating characteristic (ROC) curve of SWR detection without (orange) and with onboard

denoising (blue), alongside post hoc ripple detection (gray) ( $n_{\text{SWR}} = 2,378$ ). Paired bootstrap: online area under ROC (AUROC) 0.717 versus artificial intelligence (AI) AUROC 0.844;  $\Delta$ AUROC  $-0.127$  (95% confidence interval  $-0.162$  to  $-0.092$ ),  $P = 5.96 \times 10^{-12}$  paired bootstrap  $\Delta$ AUROC; online – AI AUROC. **e**, An example trace of CA1 LFP during theta oscillations. Scale bar, 200 ms, 0.2 mV. **f**, An example of onboard theta phase estimation compared to an offline phase based on HT. **g**, An example of optogenetic ripple closed-loop interruption. The black trace shows a detection-only trial; the blue trace shows a closed-loop optogenetic interruption trial. The gray line indicates the time of detection (scale bar, 50 ms, 0.5 mV). **h**, Firing rates of CA1 units ( $n = 7$ ) during SWRs in detection-only ( $n = 20,472$ ) and closed-loop stimulation ( $n = 10,085$ ) conditions. Data are presented as mean  $\pm$  s.e.m. **i**, WILD real-time behavior prediction. Top: an ethogram of online prediction (orange) compared to manually labeled behavior (blue). Bottom: animal trajectory during a male–male chasing event, colored by online-predicted behavior labels. Acc, accelerometer; Conv ID, one-dimensional convolution; gyro, gyroscope; MACs, multiply–accumulate operations; mag, magnetometer.

implements low-latency digital signal processing (Fig. 6a). This allows performing different operations on streamed neural signals, such as filtering in specific frequency bands and estimating the phase of an oscillation or calculating the time-varying power of a signal, such as neuronal spiking. Logical operations, such as power threshold crossing or detection of a specific oscillation phase, can be implemented and used to enable the triggering of a stimulation module. These operations can be performed on multiple neural signals simultaneously, for example, to detect synchronous events across different brain regions. Our custom algorithms enable high-accuracy signal detection with a latency of  $\sim 5 \mu$ s (Extended Data Fig. 9a–h), achieved by eliminating additional processing delays from data communications.

The other two signal processing pipelines are TinyML engines for real-time inference from pretrained ANN models for neural pattern and behavioral motif classification, respectively. These TinyML models are designed to be computationally efficient, minimizing inference time and power consumption while achieving desired accuracy. The accuracy of different neural network architectures scales differently with model size and is highly dependent on the type of input data. We found an optimal tradeoff between accuracy and latency for neural signal inference by combining a convolutional neural network (CNN),

for subsampling, and a gated recurrent unit (GRU), to capture signal temporal features (Fig. 6b and Extended Data Fig. 9b). Running this model on WILD achieved a  $>90\%$  accuracy in neural pattern detection with a latency below 10 ms (Fig. 6c).

We performed several experiments to demonstrate onboard signal processing and closed-loop stimulation capabilities of WILD. SWRs are transient (50–100 ms), high-frequency ( $\sim 150$  Hz) oscillations that occur during pauses in behavior and sleep. They have been widely studied for their important roles in learning and memory<sup>29,36</sup>. Fast and accurate neural interventions are required to investigate the causal relationship between SWR occurrence and behavioral outcomes<sup>28,31,37,38</sup>. This is notoriously difficult in behaving animals owing to the confounds introduced by other sources of high-frequency activity, such as electromyogram (EMG) activity, that contaminate hippocampal neural recordings<sup>39,40</sup>. We performed online SWR detection on a mouse with large EMG contamination (Fig. 6d). WILD TinyML model was trained to reject EMG activity, which was achieved by introducing minimal processing delay (Fig. 6b,c). After denoising, real-time SWR detection improved significantly, reaching a performance similar to post hoc detection (which uses multiple noncausal and computationally demanding processing steps) (Fig. 6d).

To demonstrate that WILD can enable experiments that are not feasible with current methods, we performed neural closed-loop manipulations in mice while conducting a spatial foraging task in a 3D maze. Mice started the trial above an elevated platform and had to find their way down and into an entrance to the structure underneath, where a food reward was hidden (Extended Data Fig. 10c). Transgenic mice expressing channelrhodopsin (ChR2) in pyramidal cells were implanted with silicon probes and optic fibers in the hippocampus and prefrontal cortex, and a WILD device with an optogenetic stimulation module. During this task, two main network activity patterns were present: SWRs, which dominated during periods of immobility, and a slower, sustained 'theta' frequency (~8 Hz) oscillation during locomotion. We performed two types of optogenetic manipulation experiments. First, we detected periods of sustained theta oscillations and computed their real-time phase; then we delivered optogenetic stimulation restricted to a specific phase (Fig. 6e,f). In control conditions, pyramidal cells were strongly phase-locked to theta oscillations but this phase-locking was disrupted by our manipulation. We also detected SWRs and disrupted them by delivering short blue-light pulses (Fig. 6g), which abolished SWRs and silenced the associated neuronal firing (Fig. 6h). This experiment demonstrates how WILD enables rapid and selective manipulation of specific neural patterns.

In another experiment, we performed real-time behavioral pattern classification in pairs of freely interacting male mice. The goal was to identify specific behavioral motifs in real time in conditions where video classification would be challenging (for example, groups of interacting mice in cluttered environments) to enable closed-loop manipulations, all in a fully wireless manner. First, we labeled different social behaviors of previously recorded videos. Then, we used these behavioral labels to train WILD's onboard TinyML model. The TinyML engine implements a behavior classifier, based on IMU signals, that can detect specific, transient behavioral motifs or 'syllables' (Fig. 6a). We obtained the best performance with a deeper GRU architecture (Extended Data Fig. 10). Finally, we used the trained model to detect different behaviors in real time. The low latency of our onboard behavioral prediction (>100 ms) allows using it to trigger optogenetic or other types of manipulations to disrupt neural activity during specific, transient behaviors, even in groups of mice during unrestricted interactions. Performance of real-time behavioral identification was highly similar to the ethogram from manually labeled video (Fig. 6i), demonstrating the accuracy of WILD onboard behavioral syllable prediction and its suitability for closed-loop manipulations.

## Discussion

We developed a modular wireless, responsive neural interface platform (WILD) that overcomes key limitations of existing technologies, enabling experiments that are not feasible with current methods for investigating the neural basis of natural behaviors in small animals. To facilitate its adoption by the community, we have made all the hardware and software elements of this platform open-source and fully documented online.

Recent advances in tethered neural interfaces have enabled large-scale neural recordings in mice and other small animals<sup>2,18,41</sup>. However, this approach is not compatible with a wide range of naturalistic behaviors, especially those involving multiple animals, large-scale spaces or cluttered environments. These limitations can be overcome using wireless devices<sup>14,15,42,43</sup>. However, existing wireless interfaces lack important capabilities such as onboard real-time signal processing or simultaneous multimodal sensing, and have limited recording bandwidth and duration under weight constraints as those imposed by freely moving small animals.

The WILD platform offers several other key advantages over previously published and commercially available systems. First, accessibility: WILD is constructed entirely from off-the-shelf components and is compatible with standard neural probes, including flexible probes, silicon

probes and tetrodes. WILD uses Bluetooth low energy (BLE) for wireless communication, allowing operations without additional expensive equipment while still supporting communication over tens of meters and millisecond-level synchronization. For applications requiring a longer range, WILD can be readily adapted to alternative wireless protocols, such as LoRa, sub-GHz radio or cellular networks. Second, functional integration: whereas existing wireless neural interfaces typically support only one recording modality (Supplementary Table 1), WILD enables concurrent or selective acquisition from a diverse set of sensors, with reserved connectivity for future expansion. Third, real-time event detection: WILD incorporates a standalone signal-processing pipeline that operates on both electrophysiological and behavioral inputs, enabling detection of neural events, as well as model-based signatures derived from neural and behavioral signals. Unlike existing systems, WILD supports concurrent closed-loop optogenetic (or electrical) stimulation and high-fidelity signal acquisition within a fully wireless architecture. These design features further enable high-performance real-time neural signal processing (NSP) and closed-loop manipulation in untethered animals, including in environments beyond conventional laboratory settings. At the same time, WILD remains more lightweight and more power-efficient than other existing wireless recording devices owing to its integrated system architecture and optimized data handling. The possibility of switching between high-power and idle recording modes on demand (for example, triggered by animal movement) enables prolonged recordings without the need for battery change. Wireless charging via photovoltaic or electromagnetic coupling<sup>44,45</sup> could also be used to extend operating time.

Compared to existing technology, our platform offers advantages in high-performance real-time NSP, enabling multimodal recordings and closed-loop neural manipulations in groups of untethered animals, even outside conventional laboratory environments. In contrast to tethered or head-fixed approaches, our platform enables circuit-level investigations of group interactions, courtship, aggression and navigation in large environments in small animals. To illustrate some potential applications, we demonstrated how mice carrying WILD devices behave in a similar way to unimplanted animals during social group behaviors, and how simultaneous recording of neural and behavioral signals uncovers previously unobserved correlates of social behaviors. Our recordings of mice living in an outdoor enclosure also enabled the observation of rodent place cells in a natural environment. This approach can thus be extended to many other types of experiments of social, spatial and other types of behaviors in a variety of animals, from rodents to birds.

Onboard ML models support real-time detection of behavioral and neural signatures. We illustrated this by performing optogenetic manipulations of specific neural patterns in mice behaving in a 3D maze, and by the real-time detection of specific behavioral motifs during social interaction. While these types of experiments have been performed before, they required tethered setups and external computers for signal processing<sup>24,31,33,46</sup>. Implementing this capability in a wireless system and integrating it with onboard optogenetic stimulation enables applying it to a much wider range of behavioral paradigms that are incompatible with tethered approaches. Furthermore, the development of embedded TinyML on our platform aligns with growing industrial interest in edge computing<sup>47</sup>. We anticipate that continued hardware advances will bring even greater computational capacity to compact processors that, deployed in neural interfaces such as WILD, can further expand their capabilities for investigating the neural mechanisms of natural behaviors both inside and outside the laboratory.

## Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41592-026-03220-9>.

## References

- Steinmetz, N. A. et al. Neuropixels 2.0: a miniaturized high-density probe for stable, long-term brain recordings. *Science* **372**, eabf4588 (2021).
- Zong, W. et al. Large-scale two-photon calcium imaging in freely moving mice. *Cell* **185**, 1240–1256.e30 (2022).
- Pisano, F. et al. Depth-resolved fiber photometry with a single tapered optical fiber implant. *Nat. Methods* **16**, 1185–1192 (2019).
- Taal, A. J. et al. Optogenetic stimulation probes with single-neuron resolution based on organic LEDs monolithically integrated on CMOS. *Nat. Electron.* **6**, 669–679 (2023).
- Mathis, A. et al. DeepLabCut: markerless pose estimation of user-defined body parts with deep learning. *Nat. Neurosci.* **21**, 1281–1289 (2018).
- Weinreb, C. et al. Keypoint-MoSeq: parsing behavior by linking point tracking to pose dynamics. *Nat. Methods* **21**, 1329–1339 (2024).
- Nourizonoz, A. et al. EthoLoop: automated closed-loop neuroethology in naturalistic environments. *Nat. Methods* **17**, 1052–1059 (2020).
- Gomez-Marin, A., Paton, J. J., Kampff, A. R., Costa, R. M. & Mainen, Z. F. Big behavioral data: psychology, ethology and the foundations of neuroscience. *Nat. Neurosci.* **17**, 1455–1462 (2014).
- Kennedy, A. The what, how, and why of naturalistic behavior. *Curr. Opin. Neurobiol.* **74**, 102549 (2022).
- Tsoar, A. et al. Large-scale navigational map in a mammal. *Proc. Natl Acad. Sci. USA* **108**, E718–E724 (2011).
- Takahashi, S., Hombe, T., Matsumoto, S., Ide, K. & Yoda, K. Head direction cells in a migratory bird prefer north. *Sci. Adv.* **8**, eabl6848 (2022).
- Kendall-Bar, J. M. et al. Brain activity of diving seals reveals short sleep cycles at depth. *Science* **380**, 260–265 (2023).
- Zhou, A. et al. A wireless and artefact-free 128-channel neuromodulation device for closed-loop stimulation and recording in non-human primates. *Nat. Biomed. Eng.* **3**, 15–26 (2019).
- Ginosar, G. et al. Locally ordered representation of 3D space in the entorhinal cortex. *Nature* **596**, 404–409 (2021).
- Forli, A., Fan, W., Qi, K. K. & Yartsev, M. M. Replay and representation dynamics in the hippocampus of freely flying bats. *Nature* **645**, 974–980 (2025).
- Jung, T. et al. A wireless subdural-contained brain–computer interface with 65,536 electrodes and 1,024 channels. *Nat. Electron.* **8**, 1272–1288 (2025).
- Lin, J., Zhu, L., Chen, W.-M., Wang, W.-C. & Han, S. Tiny machine learning: progress and futures [feature]. *IEEE Circuits Syst. Mag.* **23**, 8–34 (2023).
- Guo, C. et al. Miniscope-LFOV: a large-field-of-view, single-cell-resolution, miniature microscope for wired and wire-free imaging of neural dynamics in freely behaving animals. *Sci. Adv.* **9**, eadg3918 (2023).
- Ide, K. & Takahashi, S. A review of neurologgers for extracellular recording of neuronal activity in the brain of freely behaving wild animals. *Micromachines* **13**, 1529 (2022).
- Dennis, E. J. et al. Systems neuroscience of natural behaviors in rodents. *J. Neurosci.* **41**, 911–919 (2021).
- Zipple, M. N. et al. Competitive social feedback amplifies the role of early life contingency in male mice. *Science* **387**, 81–85 (2025).
- Egnor, S. R. & Seagraves, K. M. The contribution of ultrasonic vocalizations to mouse courtship. *Curr. Opin. Neurobiol.* **38**, 1–5 (2016).
- Ziobro, P., Woo, Y., He, Z. & Tschida, K. Midbrain neurons important for the production of mouse ultrasonic vocalizations are not required for distress calls. *Curr. Biol.* **34**, 1107–1113.e3 (2024).
- Chang, H. et al. Sleep microstructure organizes memory replay. *Nature* **637**, 1161–1169 (2025).
- Meyer, A. F., Poort, J., O’Keefe, J., Sahani, M. & Linden, J. F. A head-mounted camera system integrates detailed behavioral monitoring with multichannel electrophysiology in freely moving mice. *Neuron* **100**, 46–60.e7 (2018).
- Isaacson, M. et al. MouseGoggles: an immersive virtual reality headset for mouse neuroscience and behavior. *Nat. Methods* **22**, 380–385 (2025).
- Neunuebel, J. P., Taylor, A. L., Arthur, B. J. & Egnor, S. R. Female mice ultrasonically interact with males during courtship displays. *eLife* **4**, e06203 (2015).
- Oliva, A., Fernández-Ruiz, A., Leroy, F. & Siegelbaum, S. A. Hippocampal CA2 sharp-wave ripples reactivate and promote social memory. *Nature* **587**, 264–269 (2020).
- Buzsáki, G. Hippocampal sharp wave-ripple: a cognitive biomarker for episodic memory and planning. *Hippocampus* **25**, 1073–1188 (2015).
- Oliva, A., Fernández-Ruiz, A., Buzsáki, G. & Berényi, A. Role of hippocampal CA2 region in triggering sharp-wave ripples. *Neuron* **91**, 1342–1355 (2016).
- Fernández-Ruiz, A. et al. Long-duration hippocampal sharp wave ripples improve memory. *Science* **364**, 1082–1086 (2019).
- O’Keefe, J. & Dostrovsky, J. The hippocampus as a spatial map: preliminary evidence from unit activity in the freely-moving rat. *Brain Res.* **34**, 171–175 (1971).
- Tang, J. C. Y. et al. Dynamic behaviour restructuring mediates dopamine-dependent credit assignment. *Nature* **626**, 583–592 (2024).
- Fernandez-Ruiz, A., Oliva, A. & Chang, H. High-resolution optogenetics in space and time. *Trends Neurosci.* **45**, 854–864 (2022).
- Ouyang, W. et al. An implantable device for wireless monitoring of diverse physio-behavioral characteristics in freely behaving small animals and interacting groups. *Neuron* **112**, 1764–1777.e5 (2024).
- Joo, H. R. & Frank, L. M. The hippocampal sharp wave–ripple in memory retrieval for immediate use and consolidation. *Nat. Rev. Neurosci.* **19**, 744–757 (2018).
- Girardeau, G., Benchenane, K., Wiener, S. I., Buzsáki, G. & Zugaro, M. B. Selective suppression of hippocampal ripples impairs spatial memory. *Nat. Neurosci.* **12**, 1222–1223 (2009).
- Robinson, H. L. et al. Large sharp-wave ripples promote hippocampo–cortical memory reactivation and consolidation during sleep. *Neuron* **114**, 226–236.e6 (2026).
- Khodagholy, D., Ferrero, J. J., Park, J., Zhao, Z. & Gelinis, J. N. Large-scale, closed-loop interrogation of neural circuits underlying cognition. *Trends Neurosci.* **45**, 968–983 (2022).
- Schomburg, E. W. et al. Theta phase segregation of input-specific gamma patterns in entorhinal–hippocampal networks. *Neuron* **84**, 470–485 (2014).
- Newman, J. P. et al. ONIX: a unified open-source platform for multimodal neural recording and perturbation during naturalistic behavior. *Nat. Methods* **22**, 187–192 (2025).
- Marx, V. Neuroscientists go wireless. *Nat. Methods* **18**, 1150–1154 (2021).
- Park, S. I. et al. Soft, stretchable, fully implantable miniaturized optoelectronic systems for wireless optogenetics. *Nat. Biotechnol.* **33**, 1280–1286 (2015).
- Gutruf, P. et al. Fully implantable optoelectronic systems for battery-free, multimodal operation in neuroscience research. *Nat. Electron.* **1**, 652–660 (2018).
- Chen, J. C. et al. Self-rectifying magnetoelectric metamaterials for remote neural stimulation and motor function restoration. *Nat. Mater.* **23**, 139–146 (2024).

46. Willmore, L., Cameron, C., Yang, J., Witten, I. B. & Falkner, A. L. Behavioural and dopaminergic signatures of resilience. *Nature* **611**, 124–132 (2022).
  47. Wang, X. et al. Empowering edge intelligence: a comprehensive survey on on-device AI models. *ACM Comput. Surv.* **57**, 228:1–228:39 (2025).
  48. Yin, M. et al. Wireless neurosensor for full-spectrum electrophysiology recordings during free behavior. *Neuron* **84**, 1170–1182 (2014).
  49. Gagnon-Turcotte, G., Gagnon, L. L., Bilodeau, G. & Gosselin, B. Wireless brain computer interfaces enabling synchronized optogenetics and electrophysiology. In *2017 IEEE International Symposium on Circuits and Systems (ISCAS)* 1–4 <https://doi.org/10.1109/ISCAS.2017.8050345> (IEEE, 2017).
  50. Ji, B. et al. Recent advances in wireless epicortical and intracortical neuronal recording systems. *Sci. China Inf. Sci.* **65**, 140401 (2022).
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## Methods

### Device fabrication

Circuit design was performed in EDA software (EaglePCB, Autodesk). Printed circuit boards (PCBs) were fabricated with an online vendor (NextPCB). The device was assembled manually with a reflow station and a soldering station. Assembled devices were programmed through USB device firmware update mode and tested for power consumption and function integrity. We used a high-precision quartz oscillator for the clock (ABM11, Abracon;  $\pm 10$  ppm). The crystal load capacitor was manually trimmed to achieve  $<1$  ppm clock accuracy (tested with Time & Frequency Analyzer, Moku:Go, 0.78-ps resolution). Connectors were first protected with Kapton tape or a dummy plug; then a hermetic protection layer was formed with 3- $\mu$ m parylene-C deposition. The device was then prepared differently for standard neural connectors (A79025, Omnetics Connector Corp.) or directly bonded with a flexible neural probe. We designed the flexible bonding pattern to be the same as the neural connector landing pattern. A neural connector can be soldered to the electrode pads to be compatible with commercial silicon probes; a thin layer of low-temperature solder (SMDLTLP, Chip Quik) was first prepared on the electrode pad with a 0.1-mm stencil. The flexible neural probe was aligned to the electrode pads using a wet brush, and then prebaked at 80 °C in a constant-temperature oven for 5 min to evaporate residual water. The temperature was then raised to 160 °C for 10 min to melt the solder and form the electrical connection. The quality of the probe was checked by electrode impedance measurement in saline solution, with an Ag/AgCl electrode as a reference. A laser diode (PL450B, ams-Osram) of the opto-stimulation board was coupled to the 200- $\mu$ m optical fiber (two micro-manipulators help adjust the angle and offsets) and fixed with UV-curing optical adhesives (Optical Adhesives, Norland) when the maximal optical output position was found.

### Flexible neural probe fabrication

Flexible neural probes are fabricated in standard cleanroom facilities. Photolithography masks were designed with EDA software (L-edit, TannerEDA). Lithography of 5' Cr masks was performed with a laser writer (DWL2000, Heidelberg Instruments), followed by developing and etching (HMP900, Hamatech). The 4' silicon wafers were first cleaned with a mixture of hot sulfuric acid and hydrogen peroxide (Hot Piranha, Hamatech), and then thoroughly washed with an isopropyl alcohol, acetone and deionized (DI) water spin rinse. A 1.5- $\mu$ m layer of parylene-C was then deposited with PDS 2010 LABCOTER. The thickness of deposited parylene-C was measured by a stylus profilometer (DektakXT, Bruker). We used a lift-off process for metal patterning. Negative photoresist (PR) (AZ nlof 2020, MicroChemicals GmbH) was spun at 3,000 rpm and 30 s, followed by 110 °C soft-baking for 60 s. Photolithography was performed with a contact aligner (MA6-BA6, SUSS MicroTec), followed by 110 °C post-exposure baking for 60 s. PR was then developed with a tetramethylammonium hydroxide-based developer (AZ726MIF, MicroChemicals) for 90 s. Then, 20 nm of Ti followed by 150 nm of Au was deposited with an E-beam evaporator (Mark 50 E-beam Evaporator, CHA Industries). Metal lift-off was performed by dipping the sample in MICROPOSIT REMOVER 1165 for 24 h. Following metal lift-off, the sample was thoroughly cleaned before spin-coating a negative PR (AZ nLOF 2020, MicroChemicals GmbH) to define the Pt electrode pattern on top of the Au interconnects. A 50-nm Pt layer was then deposited using the same E-beam evaporator (Mark 50 E-beam Evaporator, CHA Industries), followed by lift-off under the same conditions as used for the Au metal pattern. Optical inspection was performed to verify the integrity of both the Au interconnects and the Pt electrode pattern. A second parylene-C layer was deposited to form full encapsulation. A thick PR mask (MEGAPOSIT SPR220, DuPont) was then spin-coated (1,500 rpm, 30 s) onto the wafer. The wafer was first soft-baked (90 °C, 30 s; 115 °C, 6 min), rested at room temperature for 20 min, and then exposed with a contact aligner. The wafers then rested for  $>1$  h, followed by post-exposure baking (90 °C, 30 s; 115 °C, 3 min),

and were subsequently put on a wafer holder to cool to room temperature, followed by development (AZ726MIF, 90 s). We used oxygen plasma etching to perform anisotropic etching of the electrode openings and probe outlines (Minilock III ICP etcher, Trion Technology). A mixture of high-conductivity PEDOT:PSS solution was prepared by mixing 10 g of PEDOT aqueous dispersion with 4 g of sorbitol. The mixture was thoroughly mixed and then sonicated for 30 min to improve homogeneity. After sonication, the solution was filtered to remove particulates. Immediately before use, 60  $\mu$ l of 1% 3-glycidyloxypropyl trimethoxysilane solution was added as a crosslinker, followed by one drop of dodecylbenzenesulfonic acid. The final solution was mixed gently before coating<sup>51</sup>. PEDOT:PSS was two-step spin-coated (500 rpm, 10 s/1,000 rpm, 60 s) on the wafer, followed by baking (115 °C, 20 min). A protective layer of PR was patterned for PEDOT:PSS patterning (AZ nlof 2020) as mentioned above, followed by oxygen plasma etching. Surface conductivity of the wafer was monitored to evaluate whether the excess PEDOT:PSS was removed. To remove the PR used for the PEDOT patterning, each sample was individually immersed in AZ NMP rinse and subjected to ultrasonic agitation at room temperature for approximately 5 s to ensure complete removal of the PR. Successful PR removal was typically indicated by a color change of the NMP rinse to yellow. The sample was then rinsed with DI water. While submerged in DI water, the surrounding Parylene-C film deposited on the wafer was carefully peeled off using fine tweezers. Subsequently, localized water flow was gently applied using a pipette along the edge of the probe to initiate delamination from the wafer. Upon release, the probe floated to the water surface. The floating probe was then carefully guided and aligned to the PCB pad using a fine brush. Owing to its thin and delicate structure, tweezers often caused tearing, so alignment was performed entirely with the brush. After alignment, the assembly was dried on a hot plate at 80 °C for approximately 10 min and then soldered to a PCB board using a low-temperature solder paste (melting point of 138 °C).

### Wireless recording with WILD and synchronization with external devices

During recording, WILD devices maintain an active Bluetooth connection with the host PC, enabling full bidirectional communication. PCs equipped with BLE modules were used to control the WILD devices. The host PC clock relies on an internal crystal oscillator, which is subject to drift over time. To minimize this, we calibrated the PC clock against an online reference using an open-access tool (Time Calibrator, Fountain Computer). A custom firmware was implemented on the WILD devices to report their internal clock values to the personal computer (PC) via serial communication, allowing us to compare device and host timestamps and quantify timing differences (Extended Data Fig. 2). Upon connection to the PC, each device underwent a 20-s clock calibration procedure to synchronize its internal clock with the host PC clock. After synchronization, the PC controlled the device through a set of commands, including initiating and stopping recordings, previewing signals, adjusting device parameters, managing external I/Os and performing impedance measurements. The source code for the WILD PC API is available via GitHub at <https://github.com/ayalab1/Neurologger>, allowing users to integrate the system with external behavioral control or tracking software. Communications from the WILD device are processed through event-driven handlers, which can be linked to the APIs of other experimental control systems. In addition, an I/O expansion board (Extended Data Fig. 2j) enables integration with external experimental hardware by providing digital input and output lines for synchronization and control. This interface allows external devices to trigger WILD operations or receive selected internal signals in real time, facilitating integration with existing behavioral control and tracking systems.

High-speed Bluetooth communication allows a single host computer to connect to up to seven WILD devices simultaneously (Fig. 5). This configuration enables real-time communication with all connected devices, including signal streaming, transmission of control

commands and other online operations. Alternatively, a larger number of devices, up to a few tens to one hundred devices, can be managed by the same host computer through sequential connections. In this mode, continuous real-time communication with all devices is not maintained, which limits certain online operations. However, during the initial connection, each device's internal clock is synchronized with the central PC clock. This synchronization ensures that recordings remain temporally aligned across devices throughout the entire recording session, even when continuous real-time communication with the host computer is not maintained.

We developed a continuous two-way BLE delay and clock-offset calibration process to align the WILD device real-time clock (RTC) with the host PC time base. During each BLE timestamp exchange, the PC recorded the packet transmission time and response arrival time, while the device recorded packet reception and response time using its onboard RTC. These timestamps were used in a Network Time Protocol-style calculation to estimate the PC-device clock offset and BLE round-trip delay. The calibration was repeated before and during recording to reduce the effect of BLE scheduling jitter and to correct residual RTC offset and drift.

Application-level BLE communication latency was estimated after clock-offset correction. The PC sent a timestamped BLE packet, and the device recorded the packet reception time using its RTC and returned this timestamp to the host. The host then calculated the communication delay from the corrected difference between PC packet transmission time and device packet reception time. This measurement reflects effective real-time BLE latency at the application level, including PC scheduling, BLE stack timing, connection interval effects, firmware processing and timestamping overhead, rather than physical-layer radio propagation delay.

External behavioral events were detected using a general-purpose I/O (GPIO)–USB interface board and timestamped directly on the host PC using the PC system time base, rather than through BLE. Because the WILD RTC was calibrated to the PC time base, PC-recorded GPIO event timestamps could be aligned with device-recorded neural, behavioral and system timestamps.

Different operations can be preprogrammed before the start of recording or controlled online through Bluetooth commands. In addition, onboard signals such as those from the IMU can be used as real-time triggers for device actions. Typical operations include remote control of recording start and stop, as well as conditional operations based on behavioral signals. For example, recording can be initiated or the device can enter sleep mode based on animal movement detected from online IMU activity. A summary table of commonly supported command operations is included in Supplementary Table 2.

WILD supports multiple communication interfaces for device configuration, electrode placement and recording. The system is equipped with GPIO lines that can be used for synchronization with external devices, control of stimulation circuits or communication with other experimental hardware. For high-bandwidth data transmission, WILD includes a USB 2.0 interface that enables real-time streaming of signals to a host computer. This interface is typically used during procedures such as electrode implantation, when high-resolution electrophysiological signals need to be monitored to guide electrode placement. To facilitate camera positioning during surgery, we provide a custom Python script that enables live streaming of the head-mounted camera images. In addition, local field potential signals from selected channels can be streamed to the host PC through the Bluetooth interface simultaneously, allowing wireless monitoring during recording sessions.

## Experimental model

All experiments conformed to guidelines established by the National Institutes of Health and have been approved by Cornell University's Institutional Animal Care and Use Committee. All rats/mice were kept in the vivarium on a 12-h light–dark cycle with ad libitum access to

food and water, except during training. Temperature and humidity in the room were kept at 68–72 °F and 40–60%, respectively. Animals were housed with a maximum of two per cage before surgery and changed to individual housing afterward. A total of 11 mice were used for electrophysiological recordings. Long-Evans rats (male and female, 3–8 months old) were used for a 128-channel electrophysiology experiment. Male B6FVBFI/J mice (3–6 months old, 27–32 g; The Jackson Laboratory) were used to compare tethered, unimplanted and wireless conditions. CaMKII $\alpha$ -Cre/+;Ai32/+ mice (approximately 25–30 g, 3–6 months of age) were used for optogenetics experiments.

## Surgical procedures

Electrophysiology implantations used flexible neural probes for mice and silicon probes for rats; for flexible neural probe implantation, mice were anesthetized with isoflurane. Skin on top of the skull was disinfected and opened with a scalpel, and the tissue on the skull was removed; then the skull was cleaned with hydrogen peroxide. Two stainless ground screw were tapped on the rear side of the lambda. A thin layer of strong dental cement (C&B Metabond, Parkell) was applied on the surface of the skull, excluding the planned craniotomy windows. A 3D-printed base plate was fixed to the skull with Metabond. Craniotomies were performed on the basis of stereotaxic coordinates of hippocampus CA1 (–2.0 mm anteroposterior, +2.0 mm mediolateral). After dura removal, flexible probes were flattened on the brain surface. A chemically sharpened 50- $\mu$ m tungsten wire (California Fine Wire) was used as an insertion tool to guide the probe for penetrating brain tissue, with a slow insertion speed (1 mm min<sup>–1</sup>). After reaching target depth, the tungsten wire was retracted, leaving the flexible probe in place. For the implantation with optical fiber, the optical fiber was inserted, followed by probe insertion with the same coordinates. Craniotomy windows were first protected with artificial dura-gel, and then covered with a 1:1 mixture of low-temperature paraffin and paraffin oil. The micro-camera was positioned toward the pupil and validated with live imaging. The device was then fixed to the base plate with dental cement. Postoperative care included administration of carprofen subcutaneously for 3 days, and the skin near the implant was treated with triple antibiotic ointment. For rat implantation, two high-density silicon probes (Diagnostic Biochips) mounted to a microdrive were used for each animal, with coordinates for bilateral hippocampus CA1 (–4.0 mm anteroposterior,  $\pm$ 3.0 mm mediolateral). Animals were placed in their home cage over a heating pad for post-surgical observation and were allowed to fully recover for >7 days post surgery. For silicon probe implantation, electrode depths were gradually adjusted until the hippocampal layers were identified with electrophysiological signatures.

## Social interaction experiment

Trials were conducted in a 31-inch, disk-shaped open-field arena. The ambient lighting in the room was adjusted to ensure uniform illumination. An overhead camera (aca1300-200uc, Basler AG; 40 fps or acA720-520 uc; 24.9 fps) was used to record videos of animal behavior. Devices were synchronized over Bluetooth before the start of the session. Video and multimodal recordings were initiated separately, with start times logged for alignment. To begin each trial, an implanted mouse was placed in the open-field arena for 5 min to familiarize itself with the environment. A second animal was then introduced to the arena and allowed to interact socially with the first animal for 15 min. Pilot sessions involving three to four animals engaging in social interactions were also conducted. Animal behavior was manually labeled on the basis of the video.

For comparison, similar behavioral recordings were conducted using a tethered system (OpenEphys or Intan Technologies Digital Acquisition Systems). These are very widely used extracellular electrophysiology systems. In this case, mice carried a pre-amplifier connected to the implanted electrode and a thin tether (–1.6-mm diameter) to bring the digitized signals to the amplifier board.

### Pupil tracking

Pupil dynamics in behaving mice were tracked using a head-mounted miniaturized camera. In the current implementation, we use a miniaturized CMOS camera with a resolution of  $320 \times 320$  pixels and a minimum focal distance of 4 mm. We characterized the spatial resolution of the camera as a function of the distance between the camera and the eye, resulting in 26.3, 50 and 83.3  $\mu\text{m}$  per pixel at distances of 4, 8 and 12 mm, respectively (Extended Data Fig. 5b). The temporal resolution of the camera recording was 16 Hz, which is sufficient to capture physiological pupil dynamics in mice<sup>52</sup>. In addition, the WILD platform can interface with a variety of other micro-cameras through standard industry interfaces such as the Serial Peripheral Interface, allowing higher-performance imaging modules to be integrated if needed.

Pupil diameter was automatically extracted using DLC (version 3.0.0). Video frames were first manually labeled to train the network. Marker points were added to track both pupil and eye contour (Extended Data Fig. 5c). Pupil metrics were then normalized by eye width, as previously reported<sup>24</sup>.

Owing to limited on-chip computational resources, complex image-processing libraries such as OpenCV are not currently supported for real-time pupil tracking on WILD.

### Outdoor recordings and analysis

Mice and rats' outdoor recordings were conducted in enclosures at one of Cornell's Field Stations. Enclosure dimensions ranged from  $6 \text{ m}^2 \times 6 \text{ m}^2$  to  $15 \text{ m}^2 \times 38 \text{ m}^2$ . Walls were stainless steel panels inserted several inches deep into the ground. Enclosures were covered with netting to prevent birds and other predators from entering. Multiple video cameras and other types of sensors (for example, radiofrequency receivers and UWB antennas) were placed in the enclosures to track animal movements. Food and water were exclusively provided in nine wooded boxes ('shelters'). Animal surgical and probe implantation procedures were identical to those of animals recorded inside the laboratory, but implants were encased in custom 3D-printed headstages. All male mice were implanted with WILD devices, which they carried for the duration of the experiment (in some cases for more than 2 weeks). Data included in this study were collected during free-range exploration (between nights 1–15), enabling chronic monitoring of neural activity and behavior in naturalistic conditions. Electrodes were implanted in the dorsal hippocampus. After surgical implants, mice were allowed to recover during a week, after that they were transferred to the field station. All animals were dropped into the field enclosures the same day, at the same time and in the center of the paddock. Researchers monitored the animals 24/7 and changed the batteries daily. Animal position and identity were continuously tracked using a custom UWB module attached to the main WILD board whose position was triangulated through antennae placed around the enclosure, providing real-time localization ( $\sim 10$  cm resolution) at  $\sim 7$ – $10$  Hz. Position data were later downsampled to 4 Hz. For trajectory visualization, a speed filter ( $>300 \text{ cm s}^{-1}$ ) was used to remove outliers, followed by smoothing based on median step size (kernel size of three samples). Trajectories were plotted as scatter plots on the aerial image of the enclosure.

Mice distance traveled was computed from UWB position data. Movement was defined as periods where instantaneous speed exceeded  $5 \text{ cm s}^{-1}$  and step distance was below 50 cm (to exclude tracking artifacts). Total distance per hour was summed and expressed in meters per hour ( $\text{m h}^{-1}$ ). Day and night classification was based on astronomical sunrise and sunset times for Ithaca, NY (42.44 N, 76.50 W), computed using the Astral library. Hour between sunset ( $\sim 19:00$ ) and sunrise ( $\sim 6:00$ ) were classified as night, the rest were classified as day. For day versus night comparisons, mean hourly distance was computed per animal for the day and night hours separately.

For place cell analysis, all putative hippocampal pyramidal cells with firing rates  $\geq 0.1$  Hz were included. Spatial rate maps were constructed by binning spiking data and occupancy into 3-cm spatial bins

and smoothing with a Gaussian kernel ( $\sigma = 3$  bins). Only periods with animal speed  $>2 \text{ cm s}^{-1}$  and occupancy  $>0.01$  s per bin were included. Spatial information was computed in bits per spike<sup>53</sup>. A neuron was classified as a place cell if it met all of the following criteria: (1) spatial information significantly above chance ( $P < 0.05$ , shuffle test), (2) at least one quality-filtered place field (contour at 20% of local peak, minimum and maximum area  $25 \text{ cm}^2$  and  $500 \text{ cm}^2$ , respectively, field firing rate  $>1.5$  mean firing rate), (3) peak firing rate  $\geq 1.0$  Hz and (4) total spike count  $\geq 50$ . Place cells were further validated using a Poisson enrichment test with Benjamini–Hochberg false discovery rate (FDR) correction ( $q < 0.05$ ) and leave-one-out robustness test ( $\geq 5$  runs, 100% pass fraction).

All quantifications for mouse data from field recordings are from the same nine animals. Statistical comparisons between day and night were performed using the two-sided Wilcoxon rank-sum test. Significant threshold was set at an alpha of 0.05. Box plots display the median and interquartile range (25–75th percentiles), and whiskers extending to 1.5 multiplied by interquartile range. Circadian plots display mean  $\pm$  s.e.m. across animals.

### 3D maze experiment

The 3D maze consisted of an elevated platform (46-cm diameter) with a movable ramp leading to a larger arena (1-m diameter). The platform had a single entrance to the empty space below it, which features 50 small holes (5-mm diameter, 3-mm deep). The task of the mice was to find the shortest path to descend the platform, find the entrance below and, once inside, find a hidden water reward. Each trial ended once the animal obtained the reward, after which it was returned to its home cage. The relative position of the descending ramp and entrance, as well as the location of the hidden reward changed every session, forcing mice to learn a new efficient trajectory. In addition, the experiment was conducted in the darkness, so animals had to rely on path integration and memory to solve it. The maze was thoroughly cleaned with 50% isopropyl alcohol between trials. Mice were implanted with silicon probes and optic fibers in the dorsal hippocampus. In some of the sessions, different optogenetic perturbations were applied ('Closed-loop optogenetic manipulations with WILD' section).

### Closed-loop optogenetic manipulations with WILD

A two-channel optogenetic stimulation module was developed to interface with the WILD platform. The module is controlled by a microcontroller (STM32G071, STMicroelectronics), which regulates stimulation parameters and communicates with the WILD system via a four-pin GPIO interface. A dual-channel current driver (TPS61151, Texas Instruments) provides independent control of two laser diodes or light-emitting diodes, with a maximum output current of 35 mA per channel. Stimulation currents are set by the microcontroller using onboard digital-to-analog converters. Key stimulation parameters, including intensity, duration, duty cycle and waveform, are configured through commands from the WILD system. Arbitrary waveforms can be easily programmed. The optogenetic module measures  $11.3 \text{ mm}^2 \times 9.5 \text{ mm}^2$  and weighs 0.6 g, including two mounted laser diodes (PL450B, Thorlabs).

WILD can be configured to deliver multiple types of optogenetic manipulations. Importantly, its advanced real-time signal processing capabilities allow using diverse types of signals, such as neural activity or IMU, to condition on-demand optogenetic stimulation (that is, 'closed-loop manipulations'). We illustrated two general case applications, such as the detection and disruption of a transient neural event (hippocampal SWRs), and a phase-specific stimulation during hippocampal theta oscillations. Theta closed-loop and ripple closed-loop experiments were conducted on separate days, with control and experimental groups tested in an interleaved manner. Before the closed-loop experiments, the laser power was adjusted on the basis of the amplitude of the evoked response. Theta closed-loop stimulation used a

lower-power laser that did not evoke a clear neural response, whereas ripple closed-loop stimulation required a higher power sufficient to elicit an identifiable evoked response that interrupted ripple events.

### Software development

The embedded software was developed in C and compiled using the ARM toolchain. The PC control interface was implemented in C# with Microsoft Visual Studio 2022, providing functions for real-time device control and data download.

### Onboard real-time signal processing with WILD

WILD embedded a signal processing pipeline and a TinyML inference engine. The signal processing pipeline consists of two parallel units for detecting biomarkers of local field potentials (LFPs). Data from the main ephys buffer is fed to the signal processing pipeline through an optional channel mixer that supports any linear combination of all recording channels. This enables operations such as noise subtraction, current-source-density, or extraction of principal component analysis or independent component analysis. An immediate anti-aliasing filter is then used to low-pass filter the signal to 500 Hz before down-sampling to 1,250 Hz. A variety of first-order bandpass filters can be selected to filter the signal to the band of interest, and then the power envelope of the signal can be extracted. Two different envelope extraction techniques can be selected: (1) rectification-based: rectification of bandpassed signal, followed by lowpass filtering, or (2) Hilbert transformer-based: a set of Hilbert transformer filters designed to introduce a  $-90^\circ$  phase shift to the band of interest from previous band-pass filters, which will form an analytic signal together with the signal before Hilbert transform (HT). Hence, the phase angle can be obtained by calculating the arctangent of the ratio between the phase-shifted and original signals, while the envelope can be derived from the magnitude of these two signals combined. A threshold that is either calculated by multiplying a coefficient of the mean baseline level or arbitrarily set through wireless commands is used for event detection. Following event detection, a TinyML-based noise identification pipeline is triggered to reject artifacts using a 0.5-s signal window preceding the event. A second TinyML engine performs behavior classification by continuously processing 1-s segments of IMU data. The identified behavior labels are logged and transmitted wirelessly to the PC, enabling flexible automation of experimental control. TinyML models are pretrained on a PC and compiled to firmware before experiments.

### Neural data analysis

Neural data analysis was conducted using MATLAB and Python. Electrophysiology data are processed with a custom MATLAB pipeline, including LFP generation, noise channel rejection, automatic sleep scoring<sup>54</sup> and spike sorting<sup>55</sup>, followed by manual curation in a Python graphical user interface<sup>56</sup>. Single-unit activity was binned in 25-ms intervals, and neuronal assemblies were identified using independent component analysis, with significance determined by null models, as implemented in a Python package (neuro\_py<sup>57</sup>). Patterns were sign-aligned, and assembly activity was computed by projecting normalized spiking activity onto these patterns<sup>58</sup>. Spectra analysis of LFPs was performed with Gabor transformation<sup>59</sup>. SWR detection was performed as follows. Candidate SWRs were required to exceed 2 s.d. above the mean power envelope and have durations between 20 ms and 600 ms to be classified as SWRs. For Theta wave analysis, LFPs were first filtered with a zero-phase shifting Butterworth filter with a passband between 4 Hz and 8 Hz, and the instantaneous phase was calculated by extracting the phase angle from the analytic signal produced by the HT. To compare the recording quality between the tethered and WILD system, same-day recordings were performed with a tethered system (RHD2000, Intan Technologies), followed by recording performed with the WILD system. Recordings were segmented according to sleep states (NREM and rapid eye movement) based on sleep scoring of hippocampus

LFP. Continuous segments longer than 10 s were selected for analysis. Power spectral density (PSD) was computed using Welch's method (512-point windows, no overlap, 625-point fast Fourier transform). Median PSDs were calculated for each state, and distributions were statistically compared across devices using two-sample *t*-tests with Bonferroni correction. Spectral correlations were quantified using Pearson correlation of log-transformed PSDs. Noise performance was assessed by comparing recordings from grounded channels in both the tethered system and the WILD system. Signals were high-pass filtered with a second-order Butterworth filter (cutoff of 1 Hz), and root mean square noise was calculated across channels. Noise levels were statistically compared using two-sample *t*-tests.

### Theta phase coherence

To quantify dynamic communication between the hippocampus and LS, we measured time-resolved theta phase synchronization<sup>60</sup>. Specifically, we extracted the instantaneous phase of the 4–12 Hz theta band via the HT and calculated the continuous phase-locking value<sup>61</sup> by averaging the complex phase differences over a 150-ms sliding window.

### Behavioral analysis

An over-the-head camera was used to record animal behaviors for indoor experiments. DLC (version 3.0.0) was used for tracking animal positions with at least three track points (nose, body and tail). After position tracking, animal behaviors were manually inspected frame by frame and classified into motif and then cross-checked by two different experts, based on previously reported standards<sup>62,63</sup>. USVs were processed with the DeepSqueak toolbox<sup>64</sup>. USVs were first automatically detected, followed by manual curations to identify all vocalization events. USV features, including contour shape, duration and frequency, were used for visualizing USV features with t-distributed stochastic neighbor embedding<sup>65</sup>. Call types were clustered with an autoencoder and USV contour. Nine-axis IMU data were processed with an AHRS filter from MATLAB. To compare the head movement of two socially interacting animals in tethered and untethered scenarios, a pair of animals was fit with a tethered headstage with three-axis accelerometers (RHD2000, Intan Technologies) or a WILD device. Principal component analysis was applied to accelerometer data to eliminate the device positioning differences among animals before pitch and roll calculation.

Mouse behaviors were manually labeled from inspection of recorded videos and categorized as social or nonsocial. Nonsocial behavior was defined as periods in which the animals were neither interacting nor directly facing one another. Social behaviors included nose sniffing (direct nose contact with the other animal's nose or face), body sniffing (nose contact with the trunk of the other animal's body) and anogenital sniffing (nose contact with the other animal's anal/genital region). Additional social behaviors included chasing (one animal running away while the other follows directly behind), approaching (one animal actively moving toward another) and looking (both animals facing one another but without nose contact). Tail rattling and fighting were not observed during recording sessions.

### Behavior manifold embedding

Behavioral manifolds were constructed from synchronized pose and IMU features, following the general logic of prior wavelet-based unsupervised behavior-mapping approaches<sup>66</sup>. Body geometry was computed from tracked landmarks and combined with z-scored IMU roll, yaw and pitch signals, with optional inclusion of inter-animal distance. These features were downsampled and transformed into 1–30 Hz time-frequency representations using continuous wavelet transforms. Wavelet amplitudes across channels and frequencies were concatenated and normalized at each time point, and pairwise Jensen–Shannon distances were computed between behavioral states. The resulting distance matrix was embedded into two dimensions with

semi-supervised UMAP, in which behavioral labels were incorporated to modestly guide the organization of the manifold while preserving the structure imposed by the feature space.

### Onboard ML

Model trainings were performed in Python. Neural and behavioral data were loaded from electrophysiology (LFP) and IMU. Raw accelerometer signals were converted to physical units (grams) and preprocessed to obtain pitch, roll and derived features, including signal magnitude and jerk. Labels were loaded from manually curated labels based on DeepLabCut (DLC) tracking. To address class imbalance, data were resampled by oversampling the minority classes to match the majority class. Labels were one-hot encoded for training. The dataset was split into training and validation sets (80/20). Model architectures were implemented in TensorFlow. The network was based on GRU layers and one-dimensional CNN, with hyperparameters optimized using grid search with cross-validation. Training was performed with categorical cross-entropy loss and Adam optimizer on a desktop PC with a graphics processing unit (Nvidia GTX 1080). The best-performing models were converted to TensorFlow Lite with post-training quantization for deployment on embedded hardware. Trained models were converted to Tensorflow-lite models and then compiled with STEdgeAI (STMicroelectronics). All analysis code was implemented in MATLAB 2020a and Python 3.9.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

### Data availability

Data are available via Zenodo at <https://doi.org/10.5281/zenodo.18879184> (ref. 67). Owing to their large size, the original binary data files that support the findings of this study are available from the corresponding authors upon request. Source data are provided with this paper.

### Code availability

Instructions, design files, source code and compiled binaries are available via GitHub at <https://github.com/ayalab1/Neurologger>. Scripts used for data analysis are available via Zenodo at <https://doi.org/10.5281/zenodo.18879184> (ref. 67). The device manual and operation instructions are available via GitHub at <https://ayalab1.github.io/Neurologger/>. This code is available open-source under gpl-3.0 license.

### References

51. Khodagholy, D. et al. Organic electronics for high-resolution electrocorticography of the human brain. *Sci. Adv.* **2**, e1601027 (2016).
52. Payne, H. L. & Raymond, J. L. Magnetic eye tracking in mice. *eLife* **6**, e29222 (2017).
53. Skaggs, W., McNaughton, B. & Gothard, K. An information-theoretic approach to deciphering the hippocampal code. In *Advances in Neural Information Processing Systems* Vol. 5 (eds Hanson, S. et al.) 1030–1037 (Morgan-Kaufmann, 1992).
54. Petersen, P. C., Siegle, J. H., Steinmetz, N. A., Mahallati, S. & Buzsáki, G. CellExplorer: a framework for visualizing and characterizing single neurons. *Neuron* **109**, 3594–3608.e2 (2021).
55. Pachitariu, M., Steinmetz, N., Kadir, S., Carandini, M. & D. H. K. Kilosort: realtime spike-sorting for extracellular electrophysiology with hundreds of channels. Preprint at *bioRxiv* <https://doi.org/10.1101/061481> (2016).
56. Rossant, C. et al. phy: interactive visualization and manual spike sorting of large-scale ephys data (Python). The Cortical Processing Laboratory at UCL. *GitHub* <http://github.com/cortex-lab/phy> (2023).

57. Harvey, R. Neuro py. *GitHub* [http://github.com/ryanharvey1/neuro\\_py](http://github.com/ryanharvey1/neuro_py) (2025).
58. Harvey, R. E., Robinson, H. L., Liu, C., Oliva, A. & Fernandez-Ruiz, A. Hippocampo–cortical circuits for selective memory encoding, routing, and replay. *Neuron* **111**, 2076–2090.e9 (2023).
59. Buckheit, J. B. & Donoho, D. L. Wavelab and reproducible research. In *Wavelets and Statistics* 55–81 (Springer, 1995).
60. Sharif, F., Tayebi, B., Buzsáki, G., Royer, S. & Fernandez-Ruiz, A. Subcircuits of deep and superficial CA1 place cells support efficient spatial coding across heterogeneous environments. *Neuron* **109**, 363–376.e6 (2021).
61. Lachaux, J.-P., Rodriguez, E., Martinerie, J. & Varela, F. J. Measuring phase synchrony in brain signals. *Hum. Brain Mapp.* **8**, 194–208 (1999).
62. Gygas, M., Fortes, M. S., Voelkl, B., Würbel, H. & Novak, J. Rattling the cage: behaviour and resource use of mice in laboratory and pet cages. *Appl. Anim. Behav. Sci.* **278**, 106381 (2024).
63. Jabarin, R., Netser, S. & Wagner, S. Beyond the three-chamber test: toward a multimodal and objective assessment of social behavior in rodents. *Mol. Autism* **13**, 41 (2022).
64. Coffey, K. R., Marx, R. E. & Neumaier, J. F. DeepSqueak: a deep learning-based system for detection and analysis of ultrasonic vocalizations. *Neuropsychopharmacology* **44**, 859–868 (2019).
65. van der Maaten, L. & Hinton, G. Visualizing data using t-SNE. *J. Mach. Learn. Res.* **9**, 2579–2605 (2008).
66. Berman, G. J., Choi, D. M., Bialek, W. & Shaevez, J. W. Mapping the stereotyped behaviour of freely moving fruit flies. *J. R. Soc. Interface* **11**, 20140672 (2014).
67. Zhao, Z. et al. A wireless modular platform for neuro-behavioral recording and closed-loop manipulation in small animals. In *A wireless modular platform for neuro-behavioral recording and closed-loop manipulation in small animals*. *Zenodo* <https://doi.org/10.5281/zenodo.18879184> (2026).

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### Author contributions

Z.Z. and J.P. fabricated the devices, Z.Z. developed the software, Z.Z., H.C., C.L., P.P., M.Q.A., A.O. and A.F.-R. performed the experiments, Z.Z., H.C. and P.P. analyzed the data, A.F.-R., Z.Z. and A.O. wrote the paper with input from all authors, and A.F.-R. and A.O. supervised the work.

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### Competing interests

The authors declare no competing interests.

### Additional information

**Extended data** is available for this paper at <https://doi.org/10.1038/s41592-026-03220-9>.

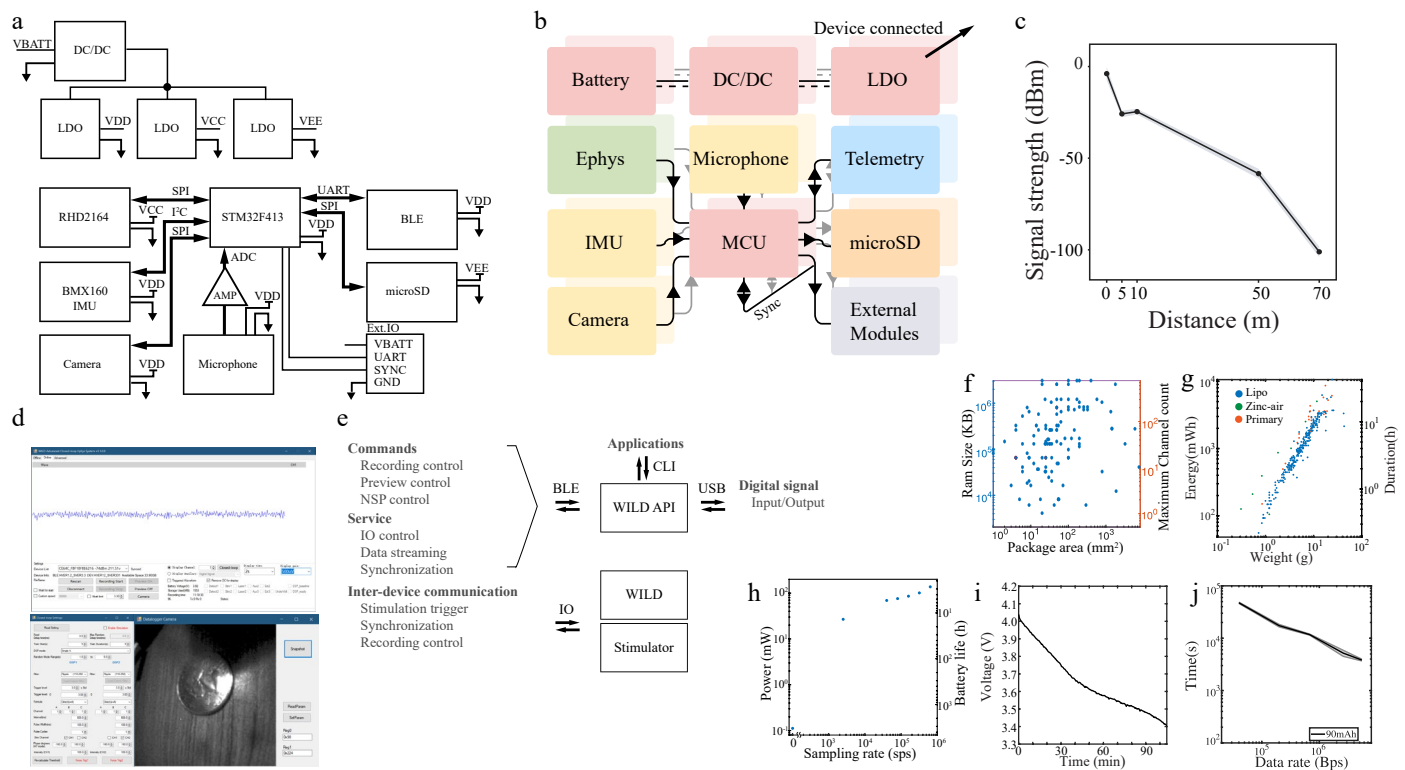
**Supplementary information** The online version contains supplementary material available at <https://doi.org/10.1038/s41592-026-03220-9>.

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### Extended Data Fig. 1 | System schematics of WILD and performance metrics.

**a** System diagram of WILD. The power system uses a high-efficiency DC-DC converter to operate across a wide input range (1.6–5.5 V), generating a 3.5 V rail. This is further regulated by three low-dropout regulators, providing isolated 3.3 V supplies for the analog system (VCC), main digital system (VDD), and peripheral digital system (VEE). The core system integrates acquisition circuits, including a neural amplifier, 9-axis IMU, camera, and ultrasonic microphone, followed by a 10× instrumentation amplifier. Processing and communication are handled by a high-performance Cortex-M4 MCU, a Bluetooth Low Energy subsystem, and a microSD card. External I/O (Ext. IO) carries data, synchronization signals, and power to connected subsystems.

**b** Function diagram of WILD subsystems.

**c** Received signal strength of the WILD Bluetooth module with the onboard antenna plotted as a function of distance. Stable communication was maintained up to 70 m between WILD and a laptop using the onboard antenna.

**d** Graphical user interface of the WILD system on a Windows PC. The top panel shows neural activity (one channel) streaming in real time, and the main control menus for recording and visualization control. The bottom left panel shows real-time event

detection and closed-loop stimulation controls. The right panel shows a video camera stream and controls.

**e** Schematic of the command pathway within the WILD ecosystem.

**f** Package area versus onboard RAM for candidate chips used in the logger design. Each point likely represents one chip. Smaller package area is better for a more compact and lightweight logger, whereas larger RAM size allows more buffering, preprocessing, or support for more recording channels.

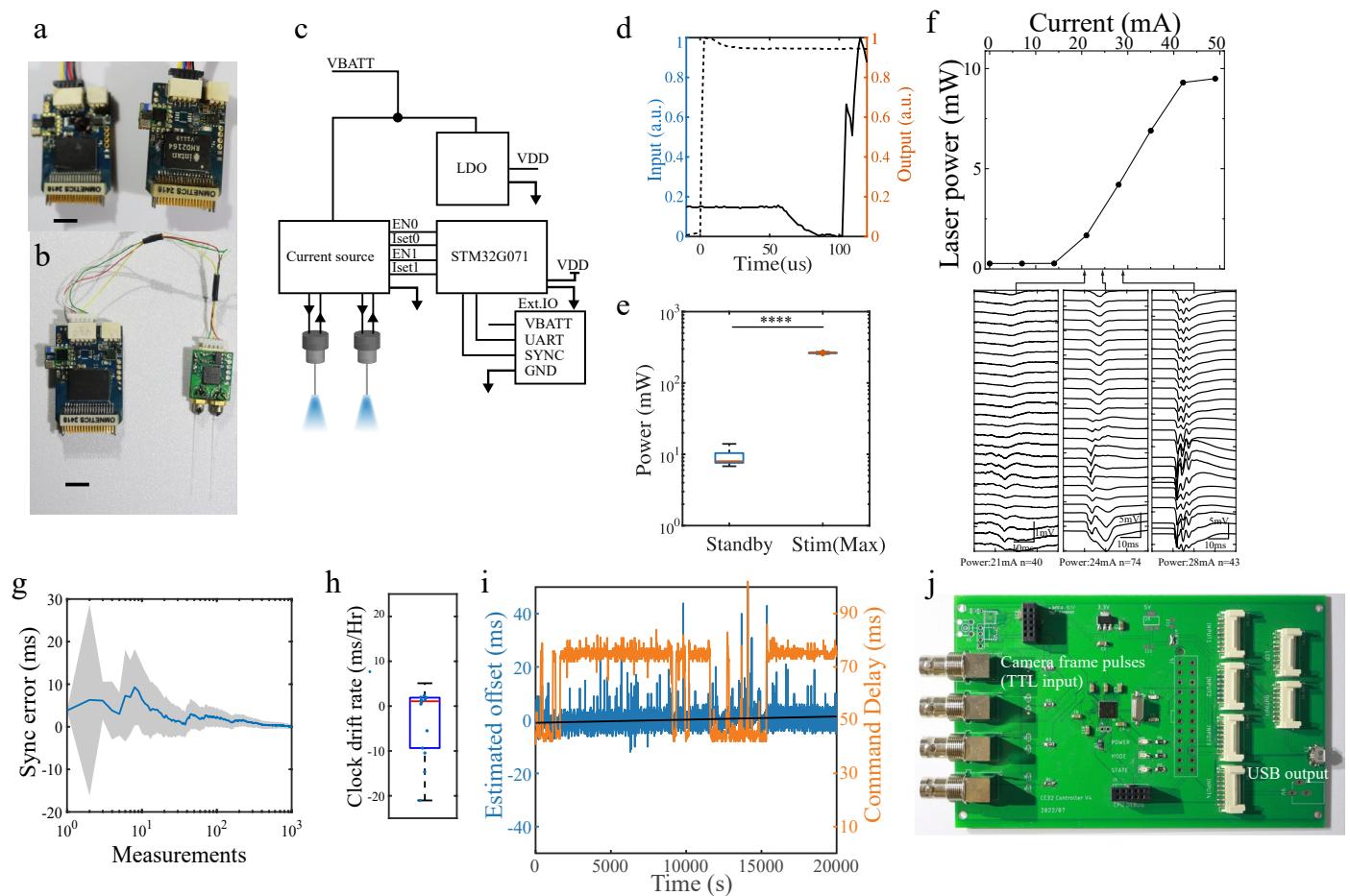
The right y axis (“minimal channel count”) converts RAM into an estimated number of channels.

**g** Comparison of weight, energy of commercially available batteries, and corresponding operation time based on high-speed recording mode (20 kHz, 64 channels) of WILD.

**h** Comparison of power consumption of the WILD system with different electrophysiology sampling rates (64 recording channels).

**i** Example battery discharge curve of WILD system powered by a 75 mAh lithium-polymer battery (64 recording channels at 20 kHz).

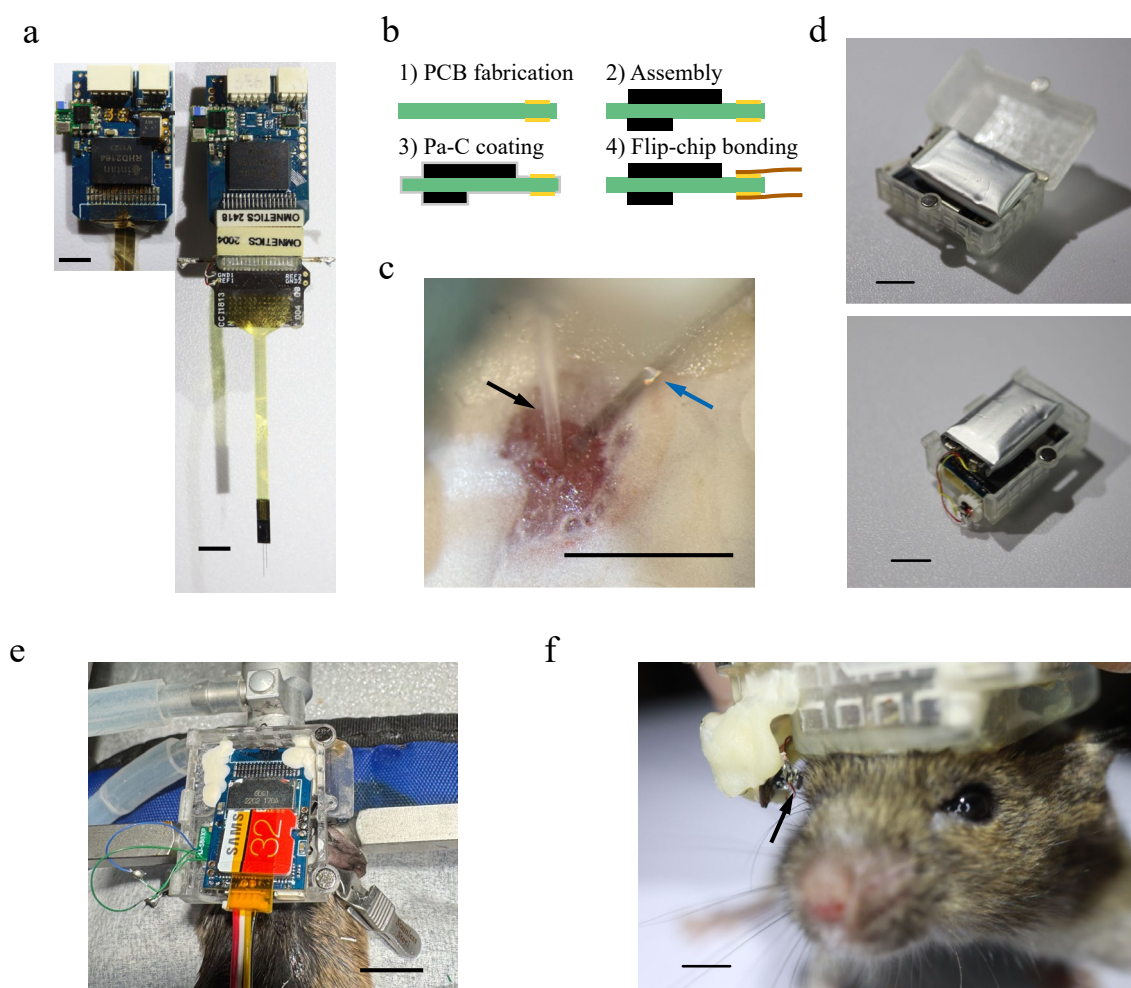
**j** Estimated device operating time as a function of data rate. Log-scaled plot showing the tradeoff between communication bandwidth and battery life for a 90 mAh power source. Higher data rates are associated with shorter operating times, illustrating the power cost of increased data throughput.



### Extended Data Fig. 2 | Wireless control of WILD with a custom framework.

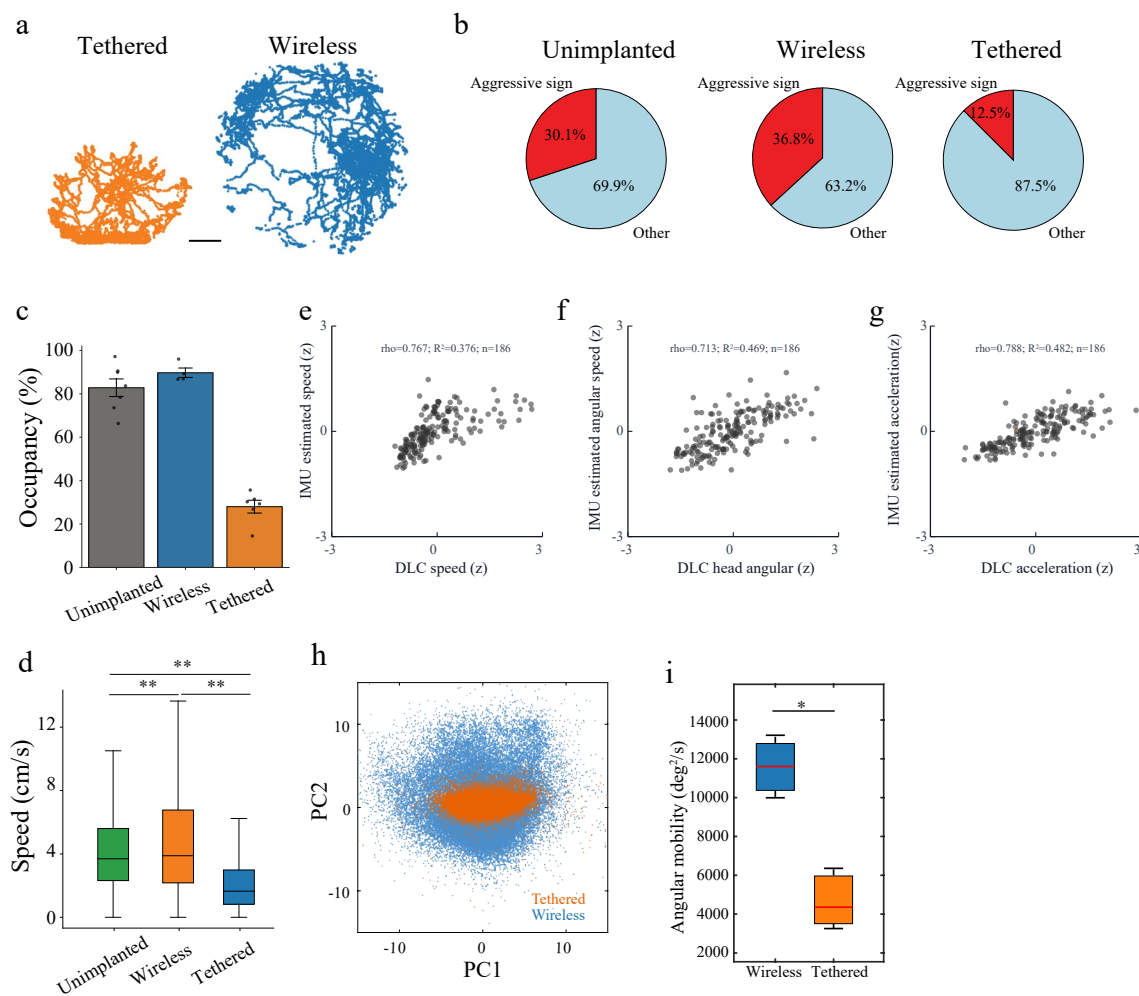
**a** Two WILD boards with standard connectors synchronized through an external IO cable (Scale bar, 5 cm). **b** WILD board connected to optogenetic stimulator subsystem through a custom 4-wire ribbon (Scale bar, 5 cm). **c** System diagram of the WILD optical stimulation module. A dual-channel current source drives two laser diodes. A compact microcontroller controls the stimulation parameters, while a low-dropout regulator (LDO) supplies power to the MCU. **d** Time-resolved relationship between the stimulation input and emitted optical output, illustrating optogenetic stimulation module response dynamics and enabling measurement of power-on latency. **e** Power consumption of the optogenetic stimulation module under standby conditions and during maximal stimulation (two-sided t-test  $p = 2.2 \times 10^{-308}$ ,  $n = 5,001$  samples). Box-plots show median and 25th and 75th percentiles, with whiskers as the minimum and maximum values. **f** WILD drives effective optogenetic stimulations. Upper panel: Blue light power measured from the tip of the optical fiber couple to the WILD optogenetic module as drive current increases. Bottom panel: Hippocampal evoked response as a result of optogenetic stimulation at increasing laser power. **g** Clock synchronization error of WILD relative to the host PC during repeated over-the-air synchronization. The shaded area represents the 95% confidence

interval. To maximize accuracy, the crystal oscillator was first calibrated by load tuning, followed by a time-of-flight-based estimation to align device time with the PC. Time estimation was then performed continuously throughout the recording to further refine synchronization. After this initial calibration, the device no longer required a continuous wireless link to the PC, allowing recordings from more animals than the typical seven-device connection limit. **h** Measured clock drift with respect to the host PC. The box shows the interquartile range (IQR, 25th-75th percentile) with the median indicated by a central line. Whiskers extend to the most extreme data points within  $1.5 \times$  IQR. The mean clock drift was  $-3.12 \pm 7.81$  ms (mean  $\pm$  s.d.,  $n = 14$ ). Note that PC clock drift can only be corrected in steps of approximately 20 ms per day. **i** Estimated clock error and wireless communication delay between the PC and WILD over time. The estimated clock error was  $0.0 \pm 4.3$  ms, and the estimated wireless communication delay was  $74.0 \pm 13.5$  ms (mean  $\pm$  s.d.,  $n = 6,185$  measurements). The black line indicates the fitted internal clock error relative to the calibrated PC clock, corresponding to an offset of  $-0.98$  ms and a drift of  $0.42$  ms/h. **j** GPIO board to synchronize external devices, such as video cameras, with WILD. For a detailed list of operations enabled see Supplementary Table 2.



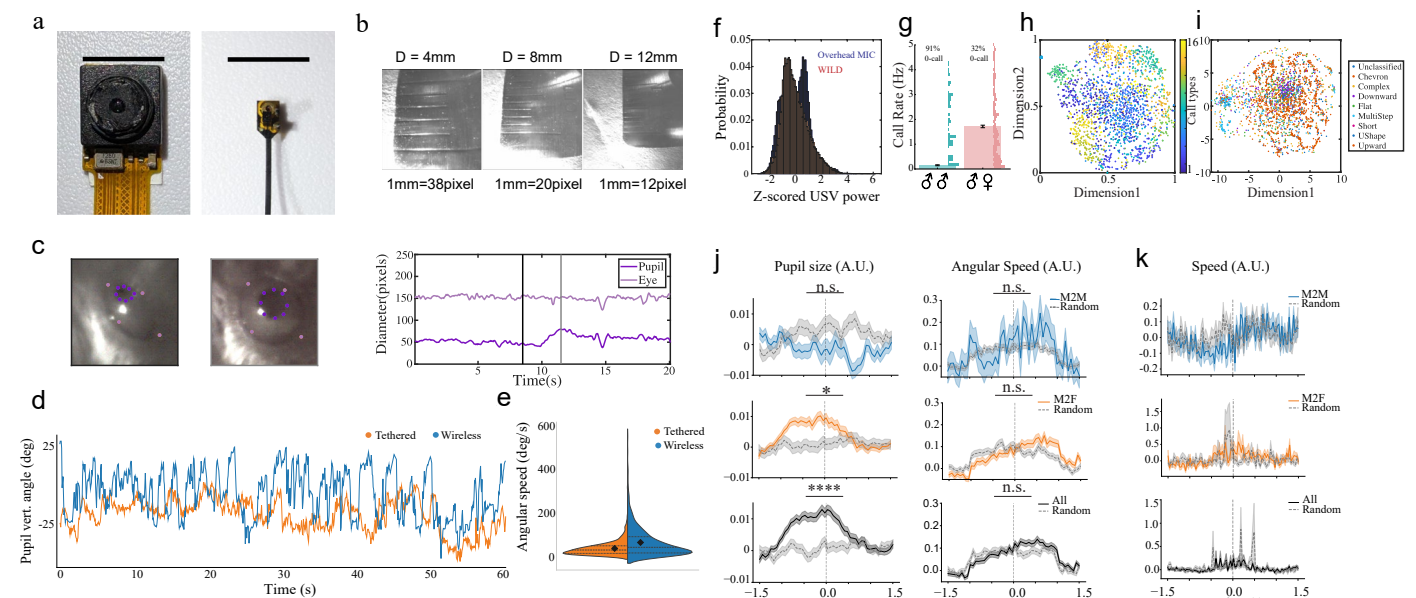
**Extended Data Fig. 3 | Integrated implantable device for closed-loop optogenetic experiments.** **a)** Left: WILD board mounted with paralyne-C flexible neural probe (Scale bar, 5 cm). Right: WILD board connected to a standard 64ch silicon probe. Scale bar, 5 cm. **b)** Illustration of the bonding process of weather-proof preparation of WILD. **c)** photo showing insertion of optical fiber (black arrow) and flexible neural probe (blue arrow) in the same implantation

craniotomy (scale bar, 1 mm) for simultaneous recording and optogenetic stimulation with WILD. **d)** Fully assembled device with custom designed 3D-printed case. Device is at the bottom, with battery on top. **e)** Intraoperative image of an animal with an implanted device, electrodes, and encasing (bottom part only) in place (scale bar, 1 cm). **f)** Photo of the pupillometry camera assembly in an implanted mouse. The arrow indicates the camera.



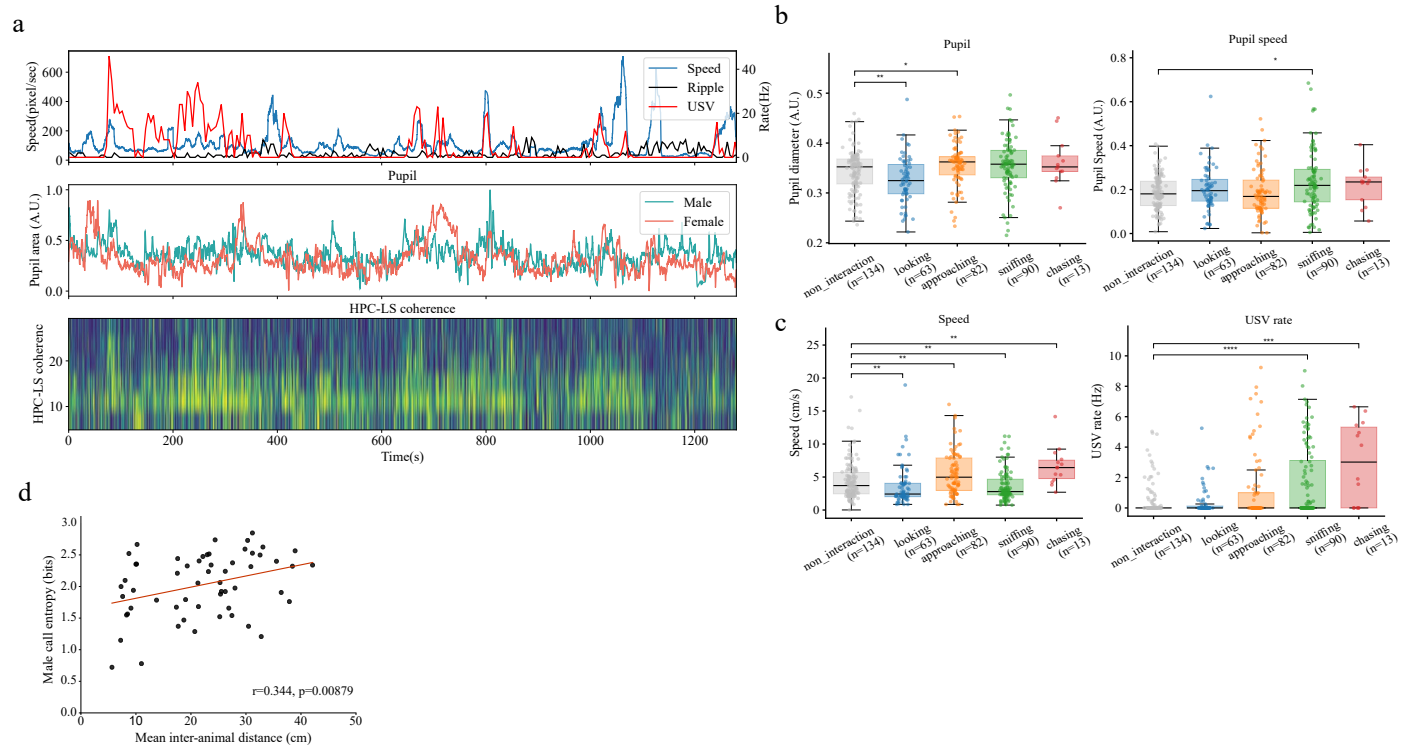
**Extended Data Fig. 4 | WILD enabled naturalistic behaviors in wireless recording settings. a)** Example tethered and wireless recorded mouse trajectories during a 15-minute open-field session (Scale bar, 5 cm). **b)** Proportion of aggressive versus other social behaviors across experimental conditions. Each pie chart represents the composition of total interaction time within one condition: Control (unimplanted), Wireless, and Tethered male-male interaction sessions. Red slices denote time spent displaying aggressive signs (for example, tail rattling, fighting, pushing), while light blue slices represent other non-aggressive behaviors. **c)** Wireless recording enhances spatial coverage during male-male social interaction. Bar plots show average occupancy coverage of unimplanted (gray,  $n = 7$  sessions from 2 mouse), tethered (orange,  $n = 6$  sessions from 2 mouse), and wireless (blue,  $n = 4$  sessions from 2 mouse) recording conditions. Unimplanted vs wireless:  $p = 0.5273$ , Mann-Whitney U test,  $p = 0.5273$ , Nemenyi-Hochberg False Discovery Rate (FDR); Implanted vs Tethered,  $p = 0.0012$ , Mann-Whitney U test,  $p = 0.0035$ , FDR; Wireless vs Tethered,  $p = 0.0095$ , Mann-Whitney U test,  $p = 0.0143$ , FDR; **d)** Moving speed distributions across experimental conditions during social interaction. Boxplots show the distribution of linear speed (cm/s) for unimplanted, wireless, and tethered animals ( $n = 10$  sessions). The centerline indicates the median, the box bounds indicate the 25th and 75th percentiles, and whiskers indicate the minimum and maximum values. Pairwise comparisons after correction for multiple comparisons: Wireless vs Unimplanted:  $p = 0.775$

Nemenyi-Hochberg FDR; Tethered vs Unimplanted:  $p = 0.0005$  FDR; Tethered vs Wireless:  $p = 0.0025$  FDR **e)** Held-out prediction of locomotion speed from IMU features using the compact gravity-constrained local projection model. Speed was independently derived from overhead DeepLabCut tracking. Pooled performance was Spearman's  $\rho = 0.767$ ,  $R^2 = 0.376$ ,  $p = 1 \times 10^{-4}$ ,  $n = 186$  windows from 3 animals. **f)** Held-out prediction of head angular speed from IMU features using the gravity-constrained generalized linear model. Head angular speed was independently derived from DeepLabCut-estimated heading. Pooled performance was Spearman's  $\rho = 0.713$ ,  $R^2 = 0.469$ ,  $p = 1 \times 10^{-4}$ ,  $n = 186$  windows from 3 animals. **g)** Held-out prediction of locomotion acceleration from IMU features using the compact gravity-constrained local projection model. Acceleration was independently derived from overhead DeepLabCut tracking. Pooled performance was Spearman's  $\rho = 0.788$ ,  $R^2 = 0.482$ ,  $p = 1 \times 10^{-4}$ ,  $n = 186$  windows from 3 animals. For **g**–**i**, each point represents one non-overlapping 20-s window from one of three validation sessions; colors denote sessions. Box-plots show median and 25<sup>th</sup> and 75<sup>th</sup> percentiles, with whiskers as the minimum and maximum values. Variables were z-scored within session before model fitting and prediction. Axes show z-scores from -3 to +3 for visualization. Performance was evaluated using leave-one-session-out cross-validation and computed across all windows. **h)** Comparison of principal components of animal acceleration between wireless and tethered recordings. **i)** Comparison of angular mobility between wireless and tethered animals ( $n = 3$ ). Statistical comparisons were performed using two-sided Kolmogorov-Smirnov tests ( $p = 0.032$ ).



**Extended Data Fig. 5 | WILD enables pupillometry and USV recordings in behaving mice.** **a** Size comparison between the Raspberry Pi camera used in previous studies<sup>24</sup> for tethered mice recordings and the WILD camera module. Scale bar, 10 mm. **b** Head-mounted camera performance at different working distances. Representative images acquired at object distances of 4, 8 and 12 millimeter, respectively, showing reduced effective spatial resolution at longer distances. **c** Left, example of pupil and eye tracking with DeepLabCut; right, diameter of the eye (orange) and the pupil (blue) over a 20 s time window. **d** Pupil vertical position over time. Blue trace: wireless; orange trace: tethered. **e** Split violin plot showing the distribution of vertical angular speed in tethered (blue) versus wireless (orange) conditions. Each half violin represents the probability density of angular speed across 60 s of behavior, with black diamonds indicating the mean. The wireless method allowed for a broader and more variable range of eye movements compared to the more restricted distribution seen in the tethered condition (Mann-Whitney test:  $U = 362,246$ ,  $p = 8.45 \times 10^{-16}$ ). **f** comparison of USV power simultaneously detected by overhead external and head-mounted microphones. Bimodal distribution in overhead microphone denote USV power dependency animal location. **g** Ultrasonic vocalization (USV) call rates during male-male and male-female interactions. Male-female interactions elicited significantly higher USV call rates than male-male

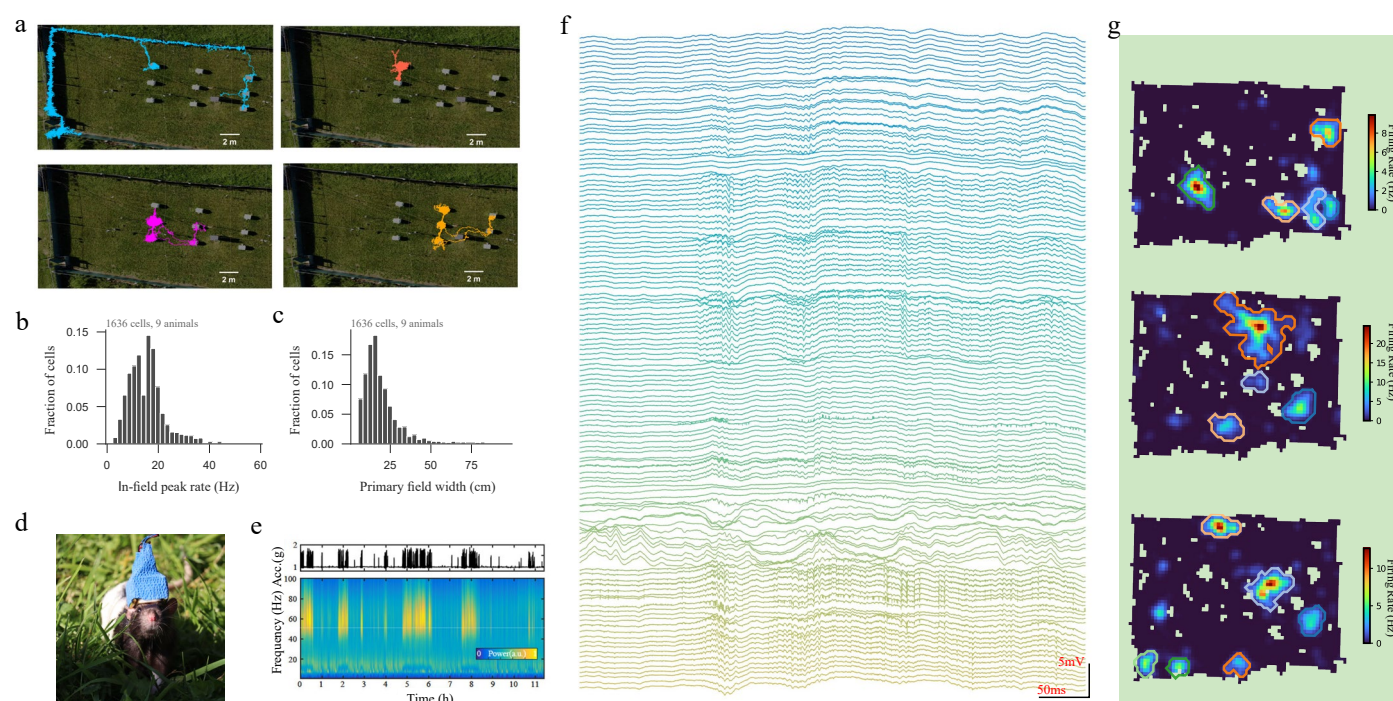
interactions ( $P = 5.5 \times 10^{-155}$ , two-sided t-test;  $n = 3$  sessions per condition). Bars show mean  $\pm$  SEM. Histograms adjacent to the box plots show the distribution of call rates. A total of 155 calls from male-male sessions and 3,032 calls from male-female sessions were detected. **h** t-SNE plot comparing call signatures (spectra, duration) between different animal call types ( $N = 2,092$ ). **i** t-SNE plot of USV spectrograms classified using DeepSqueak. Each point represents one vocalization color-coded by call type. **j** Normalized pupil size and head angular speed aligned to male vocal calls, separated by interaction target. Top: Pupil size (A.U.) aligned to male-to-male (M2M, left) and male-to-female (M2F, right) calls. M2M: not significant ( $p = 0.180$ ); M2F:  $p < 0.05$  ( $p = 0.024$ ). Bottom: Corresponding head angular speed (A.U.) around calls. M2M: not significant ( $p = 0.880$ ); M2F: not significant ( $p = 0.078$ ) (mean  $\pm$  SEM; male-to-male 155 calls from 3 sessions, male-to-female 3,032 calls from 3 sessions). **k** Normalized head speed aligned to male vocal calls across different interaction contexts. Top: Male-to-male (M2M) interactions. Middle: Male-to-female (M2F) interactions. Solid colored lines show mean normalized speed around call onset ( $t = 0$  s), with shaded areas indicating  $\pm$ SEM. Dashed gray lines represent randomized control time points (mean  $\pm$  SEM;  $P > 0.05$  across all comparisons, paired t-test). Vertical dashed line marks call onset. (M2M: 155 calls from 3 sessions, M2F: 3,032 calls from 3 sessions).



#### Extended Data Fig. 6 | Physiological correlates of different social behaviors.

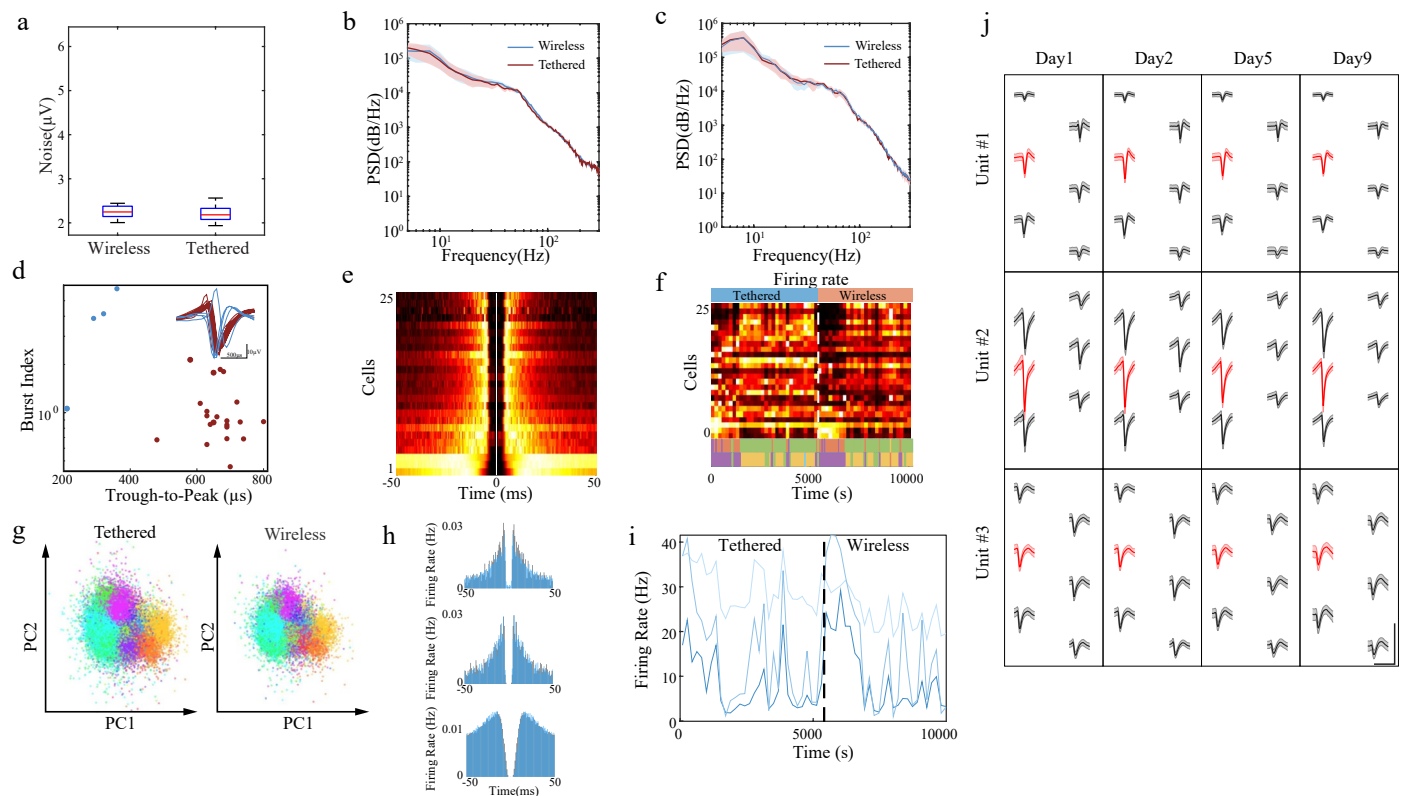
**a** Top, example traces of ultrasound vocalization signals (USV, red), animal speed (speed, blue) and sharp-wave ripples (SWRs, black) rate. Middle, simultaneous recordings of pupil size in a male (blue) and a female (orange) during unrestricted interactions. Bottom, hippocampus-lateral septum (HPC-LS) spectral LFP during the same interaction window. **b** Distribution of pupil diameter (left) and speed (right) during different social behaviors including looking ( $n = 63$  events), approaching ( $n = 82$  events), sniffing ( $n = 90$  events), chasing ( $n = 13$  events) and no interaction ( $n = 134$  events). Note that pupil diameter significantly shrank and significantly dilated during looking ( $p = 0.0065$ ) and approaching ( $p = 0.022$ ) behaviors respectively compared to no interaction, while pupil speed was higher during sniffing compared to no interaction ( $p = 0.013$ ). **c** Animal speed (left) and USV rate (right) during different behaviors. Note that animal's speed was significantly lower during looking ( $p = 0.0026$ ) and sniffing ( $p = 0.0079$ ), but

significantly higher during approaching ( $p = 0.0042$ ) and chasing behaviors ( $p = 0.0010$ ) respectively, compared to no interactions. USV rate was significantly higher during sniffing ( $p = 6.59 \times 10^{-5}$ ) and chasing ( $p = 1.6 \times 10^{-4}$ ) compared to no interactions. All tests are Mann-Whitney U tests. **c** Relationship between inter-animal distance (male and female) and entropy of the male emitted USV calls ( $n = 1683$  calls over 20 s time windows) displayed a positive relationship, red line shows linear fit ( $r = 0.3434$ ,  $p = 0.00879$ ,  $n = 57$  time windows from 2 sessions). For each 20 s window, male call entropy was calculated as the Shannon entropy of the distribution of different call types and plotted against the corresponding mean inter-animal distance. **d** Male USV call entropy increased with inter-animal distance, indicating more versatile call patterns when animals were physically farther apart (linear regression,  $r = 0.34$ ,  $p = 0.0088$ ,  $n = 57$ ). Box-plots show median and 25<sup>th</sup> and 75<sup>th</sup> percentiles, with whiskers as the minimum and maximum values.



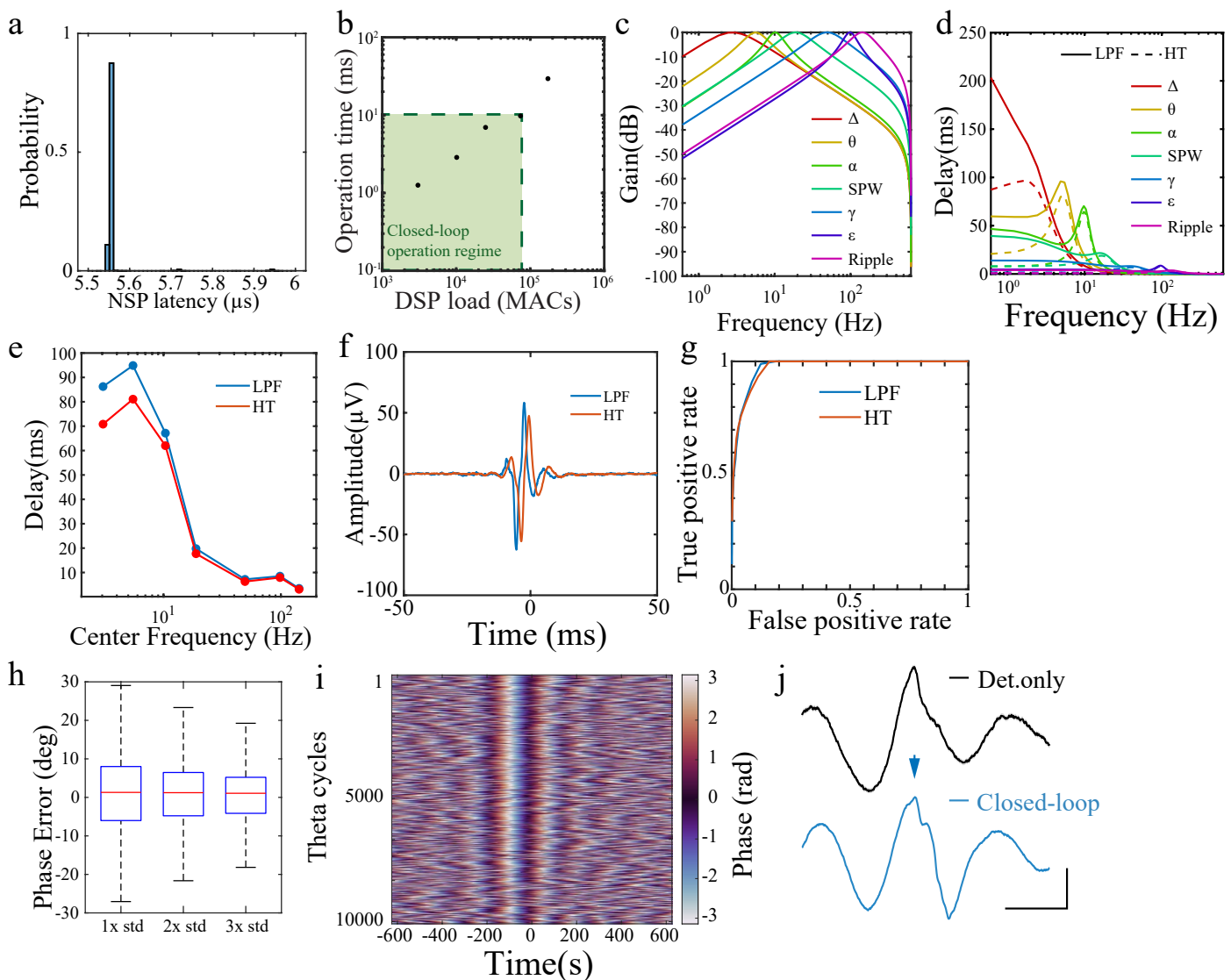
**Extended Data Fig. 7 | Individual animal trajectories and place cell population metrics.** **a** Examples of individual UWB-tracked trajectories in outdoor enclosure for four simultaneously recorded mice on night 6, plotted over the aerial image of the field enclosure. Each panel shows a different animal (color coded as in Fig. 5). **b** Distribution of primary place field width (cm) for all identified place cells (n = 1,636, n = 9 animals). **c** Distribution of in-field peak firing rate for all identified place cells (n = 1,636, n = 9 animals). **d** Image of a

rat with implanted WILD devices and silicon probes. **e** Long duration (~12 h) recording at low sampling rate (1250 Hz) of a rat implanted with WILD. **f** Large-scale (128 channels) high sampling rate (20,000 Hz) outdoor recording in a rat implanted with bilateral silicon probes and WILD. **g** Examples of three dorsal hippocampal place cells from rats exploring an outdoor enclosure (2.4 × 3.4 m) obtained after recording with WILD.



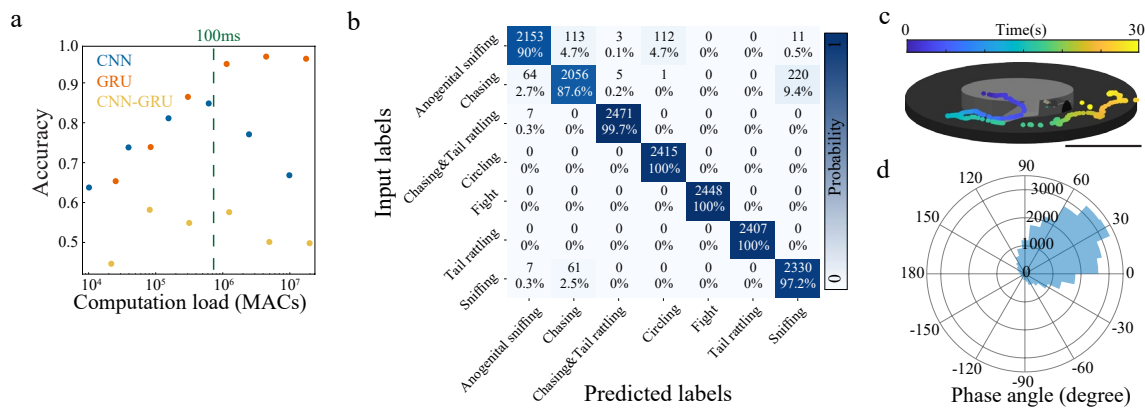
**Extended Data Fig. 8 | WILD enabled low-noise and high-quality single-unit electrophysiological recordings.** **a**) Comparison of root-mean-square noise of grounded channels between the WILD and Intan RHD2000 systems.  $V_{\text{WILD}} = 2.92 \pm 1.12 \mu\text{V}$  ( $n = 32$  channels) and  $V_{\text{Intan}} = 2.80 \pm 1.95 \mu\text{V}$  ( $n = 32$ ); paired t-test,  $p = 0.757$ . Box-plots show median and 25<sup>th</sup> and 75<sup>th</sup> percentiles, with whiskers as the minimum and maximum values. **b**) Comparison of CA1 LFP power spectral density (PSD) between WILD and Intan RHD2000 system, during NREM sleep ( $n = 23$  channels). **c**) Comparison of CA1 LFP power spectral density (PSD) between WILD and Intan RHD2000 system, during REM sleep ( $n = 23$  channels). **d**) Comparison of burst index and trough-to-peak ratio of CA1 units ( $n = 25$ )

recorded with flexible probes and WILD in behaving mice. **e**) Stacked ACGs of recorded CA1 units ( $n = 25$ ). **f**) Firing rate of CA1 units vs time. The dashed line indicates the switch point from tethered recording to wireless recording. **g**) Comparison of the first and second principal components of spike waveforms from recorded units between tethered and wireless recordings. **h**) Selected ACGs of CA1 units during wireless (blue) and tethered (gray) recordings. **i**) Firing rates of selected units during tethered and wireless recordings. **j**) Examples of waveforms from stably recorded single units with a flexible neural probe over 9 days (Scale bar, 150 ms, 50  $\mu\text{V}$ ). Peak-amplitude channels are highlighted in red, and the shaded areas represent the standard error.



**Extended Data Fig. 9 | WILD embeds a fast, accurate online neural signal processor.** **a**) Signal processing latency of the embedded neural signal processor. Delay =  $5.56 \pm 0.09 \mu$ s,  $N = 387,561$ . **b**) WILD DSP operation time as a function of DSP load across different TinyML model sizes. The green box indicates the time requirement of closed-loop neural stimulation. **c**) Frequency responses of bandpass filters used in neural signature detection. Filters are first-order Butterworth filters with different passbands: Delta( $\Delta$ ): 1–4 Hz; Theta( $\theta$ ): 4–8 Hz; Alpha( $\alpha$ ): 8–13 Hz; Gamma( $\gamma$ ): 30–80 Hz; Epsilon( $\varepsilon$ ): 80–120 Hz; Sharp-wave (SPW): 12–30 Hz; Ripple: 100–200 Hz. **d**) Signal processing delays of NSP, showing Hilbert transformer-based NSP (HT, dashed line) have reduced delay compared to lowpass-filter-based NSP. **e**) Comparison between NSP delays based on LPF mode and HT mode across bandpass filters with different central frequencies. **f**) Trigger-averaged waveforms of CA1 ripple detections obtained using HT and

LPF neural signal processing (NSP). **g**) ROC curves of ripple detection obtained using HT and LPF-based NSP, compared to a curated offline ripple detector. Area under curve: HT: 0.97218,  $n = 2,864$ ; LPF: 0.97422,  $n = 2,075$ . **h**) Hippocampus CA1 theta phase decoding error of NSP phase decoder compared with offline decoder with Hilbert Transform, with signal amplitude in Theta band exceeding  $1 \times / 2 \times / 3 \times$  standard deviation (STD,  $n_{1 \times \text{STD}} = 680,579$  samples;  $n_{2 \times \text{STD}} = 424,867$ ;  $n_{3 \times \text{STD}} = 161,774$ ). Box-plots show median and 25<sup>th</sup> and 75<sup>th</sup> percentiles, with whiskers as the minimum and maximum values. **i**) Trigger-averaged theta phase of phase-based NSP detections (phase window set to  $-15^\circ$  to  $15^\circ$ ,  $n = 10,085$ ). **j**) Triggered average of LFPs for onboard-detected events during detection-only ( $N = 1,670$ ) and closed-loop theta-wave intervention ( $n = 6,052$ ) conditions (scale bar: 100 ms, 100  $\mu$ V). Stimulation onset is indicated by the blue arrow.



**Extended Data Fig. 10 | TinyML allows rapid computations for online processing during behavior.** **a**) Inference accuracy of different TinyML models on behavior identification compared with model size and model structures. Target inference times are indicated by a dashed line. **b**) Confusion matrix comparing onboard-predicted behavior labels with ground-truth reference labels (Behavioral events were annotated and quantified, including sniffing

( $n = 194$ ), anogenital sniffing ( $n = 19$ ), chasing ( $n = 108$ ), chasing & anogenital sniffing ( $n = 3$ ), chasing & tail rattling ( $n = 11$ ), circling ( $n = 7$ ), fighting ( $n = 6$ ), pushing ( $n = 4$ ), and tail rattling ( $n = 56$ ). **c**) Example of mouse trajectory during exploration of double-layer maze (scale bar: 50 cm). **d**) Histogram of real-time stimulation theta phases ( $N = 15,728$ ). Target was 30–45°.

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|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data collection | <p>Data acquisition and retrieval:All onboard data streams, including electrophysiology, pupillometry, microphone, and motion were recorded to a microSD card during experiments and later downloaded via a custom WILD API for offline analysis.</p> <p>Electrophysiology:Neural signals were acquired using an electrophysiology amplifier chip (Intan Technologies, RHD2164) and stored in binary format compatible with the Intan RHD2000 system (<a href="https://github.com/ayalab1/Neurologger/blob/main/docs/file-format.md">https://github.com/ayalab1/Neurologger/blob/main/docs/file-format.md</a>). Recordings from a commercial RHD2000 system (Intan Technologies) were also used for validation.</p> <p>Pupillometry:Video data from the onboard Naneye-C camera were stored as continuous binary pixel streams and converted to MP4 format using a custom Python script (WILD_VideoDecodewAudio.py). For animals implanted with a Raspberry Pi camera, image acquisition was performed using an independent custom script (<a href="https://github.com/amymarple/pupil_closed_loop">https://github.com/amymarple/pupil_closed_loop</a>).</p> <p>Microphone:Acoustic signals from the onboard microphone were sampled by the microcontroller's ADC and stored as continuous audio data.</p> <p>Motion:Inertial data from the 9-axis IMU were continuously logged by the microcontroller to the same storage medium. In outdoor experiments, an ultra wide bandwidth wireless position system were used to acquire animal positions(<a href="https://wisersystems.com">https://wisersystems.com</a>). In indoor experiments, a roof-mounted camera (Basler AG) were used and video were recorded through a image acquisition software(<a href="https://www.baslerweb.com/en-us/software/pylon">https://www.baslerweb.com/en-us/software/pylon</a>).</p> <p>Laser power: A optical power meter were used to measure laser power from the optical fiber tip (Thor Labs, S130C)</p> <p>Device electrical characterization including power consumption, synchronization, processing time were measured through a combination of customized firmware, debugging interface (ST microelectronics, ST-link) and source measurement unit (Keysight, U2722A).</p> |
| Data analysis   | <p>Custom Matlab and Python scripts were used for data analysis: <a href="https://zenodo.org/record/1887918">10.5281/zenodo.1887918</a></p> <p>Spike sorting: Single units were detected and clustered using Kilosort(<a href="https://github.com/MouseLand/Kilosort">https://github.com/MouseLand/Kilosort</a>), then manually curated with PHY(<a href="https://github.com/cortex-lab/phy">https://github.com/cortex-lab/phy</a>).</p> <p>Animal position and pupil size tracking: Videos were analyzed using DeepLabCut(<a href="https://github.com/DeepLabCut/DeepLabCut">https://github.com/DeepLabCut/DeepLabCut</a>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Ultrasonic vocalization: Audios were analyzed using DeepSqueak (<https://github.com/DrCoffey/DeepSqueak>)

Model training and deployment: Machine learning models were trained with custom scripts with tensorflow(<https://github.com/tensorflow/tensorflow>) then convert to binary executable code with STEdge-AI(<https://www.st.com/en/development-tools/stedgeai-core.html>).

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All data are available from the corresponding authors upon request. Source data are available online on Zenodo (doi: 10.5281/zenodo.18879184). Code and design files are available to download at <https://ayalab1.github.io/Neurologger/>

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|                         |                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Laboratory animals      | Long-Evans rats were used for a 128-channel electrophysiology experiment. Male B6FVB/F1/J mice (3–6 months old, 27–32g; The Jackson Laboratory) were used to compare tethered, unimplanted, and wireless conditions. CaMKII $\alpha$ -Cre/+;Ai32/+ mice (approximately 25–30g, 3–6 months of age) were used for optogenetics experiments. |
| Wild animals            | N/A                                                                                                                                                                                                                                                                                                                                       |
| Reporting on sex        | Male animals were used in most experiments. In social interaction paradigms, both implanted and unimplanted female mice were introduced to facilitate social interactions with male subjects.                                                                                                                                             |
| Field-collected samples | N/A                                                                                                                                                                                                                                                                                                                                       |
| Ethics oversight        | All experiments conformed to guidelines established by the National Institutes of Health and have been approved by Cornell University's Institutional Animal Care and Use Committee.                                                                                                                                                      |

Note that full information on the approval of the study protocol must also be provided in the manuscript.

## Plants

|                       |     |
|-----------------------|-----|
| Seed stocks           | N/A |
| Novel plant genotypes | N/A |
| Authentication        | N/A |